Sensation and Perception

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AHUL261

Defining Sensation and Perception

- Sensation
  - The detection of physical energy emitted or reflected by physical objects.
  - It occurs when energy in the external environment or the body stimulates receptors in the sense organs.

- Perception
  - The process by which the brain organizes and interprets sensory information.

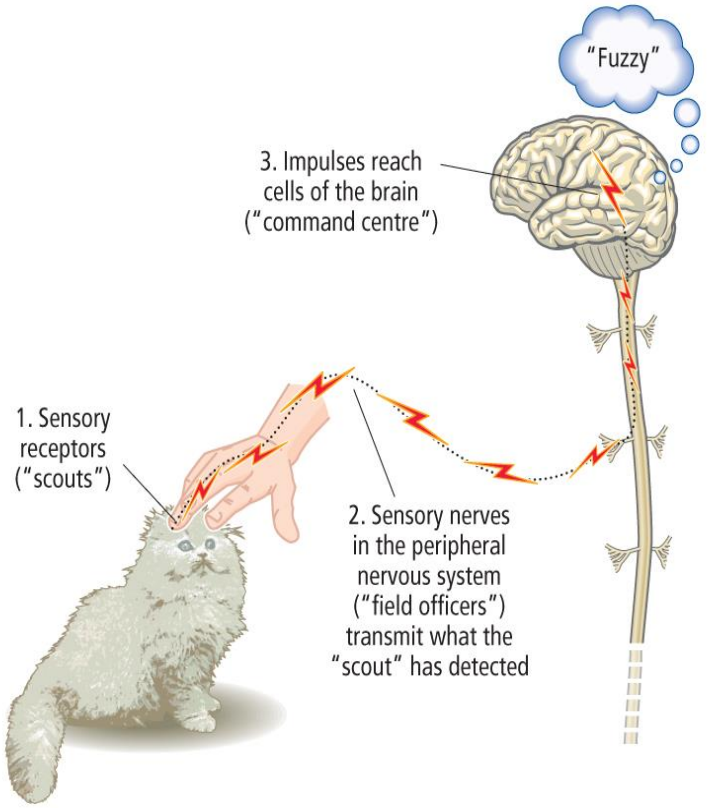
Defining Sensation and Perception

e.g. Your classmate is poking you. You first sense a poking (sensation), then you immediately know that your classmate is poking you (perception).

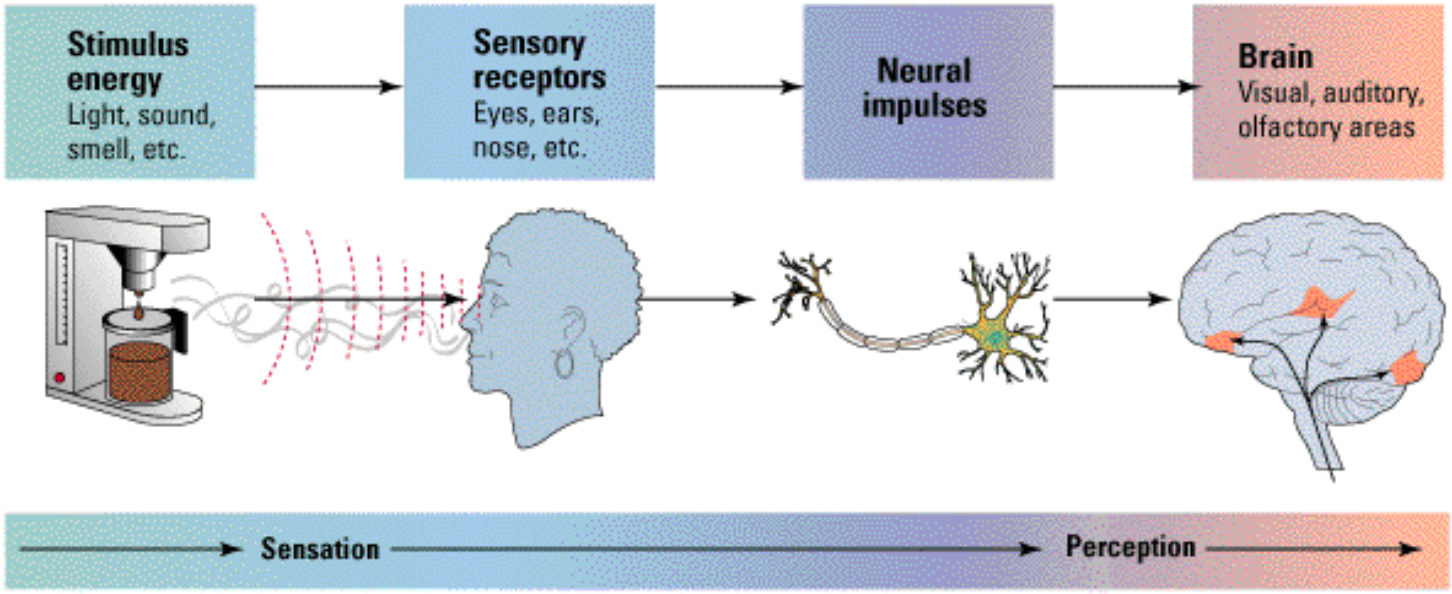
Recall Psychophysics – the study of relationship between the physical aspects of stimuli and our psychological experience of them.

Sensation

- Sense receptors
  - Specialized cells that convert physical energy in the environment or the body to electrical energy that can be transmitted as nerve impulses to the brain.



Sensation & Perception Processes



Measuring Senses

Absolute Threshold

- The smallest intensity of a stimulus that must be present for stimulus to be detected.

Absolute Sensory Thresholds

- Vision:
  - A single candle flame from 30 miles on a dark, clear night
- Hearing:
  - The tick of a watch from 20 feet in total quiet
- Smell:
  - 1 drop of perfume in a 6-room apartment
- Touch:
  - The wing of a bee on your cheek, dropped from 1 cm
- Taste:
  - 1 tsp. Sugar in 2 gal. water

Difference Threshold

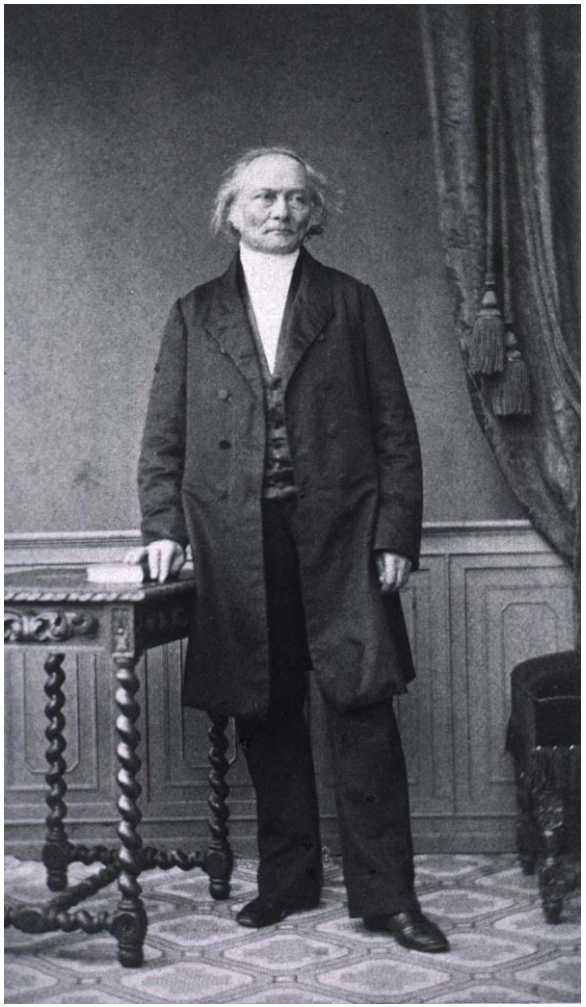
- The smallest difference in stimulation that can be reliably detected by an observer when two stimuli are compared;

- Also called Just Noticeable Difference (JND).

Weber's law

Weber's law:

A basic law of psychophysics stating that a just noticeable difference is a constant proportion to the intensity of an initial stimulus (rather than a constant amount).



Weber's Law

- The amount of change that is necessary to be noticed is systematically related to the intensity of the original stimulus
- The stronger the initial stimulus, the greater a change must be for it to be noticed.
- Mathematically:

$$K = \frac{\Delta i}{I}$$

  - *K = A constant (varies across senses)*
  - *Δi = The minimal change in the intensity required to produce j.n.d.*
  - *I = the intensity of the stimulus where the change occurs*

Sensory Adaptation

- Adaptation
  - The reduction or disappearance of sensory responsiveness when stimulation is unchanging or repetitious.

e.g. The bath that was almost too hot to enter soon feels too cool; similarly, the cold lake we jump into for a summer swim feels freezing at first, but soon feels only refreshingly cool.

Sensory Adaptation

- Advantages and Disadvantages of Sensory Adaptation
  - Prevents us from having to continuously respond to unimportant information, e.g. traffic noise
  - Might hamper response to important sensory input such as the smell of smoke. The smell of smoke might not alert you if your house is on fire.

Types of Senses: An Overview

- Visual (Vision)
- Auditory (Hearing)
- Gustatory (Taste)
- Olfactory (Smell)
- Tactile (Touch)
- Vestibular (Balance)
- Kinesthetic (Posture & movement)
- Organic senses (internal organs & physiological processes)

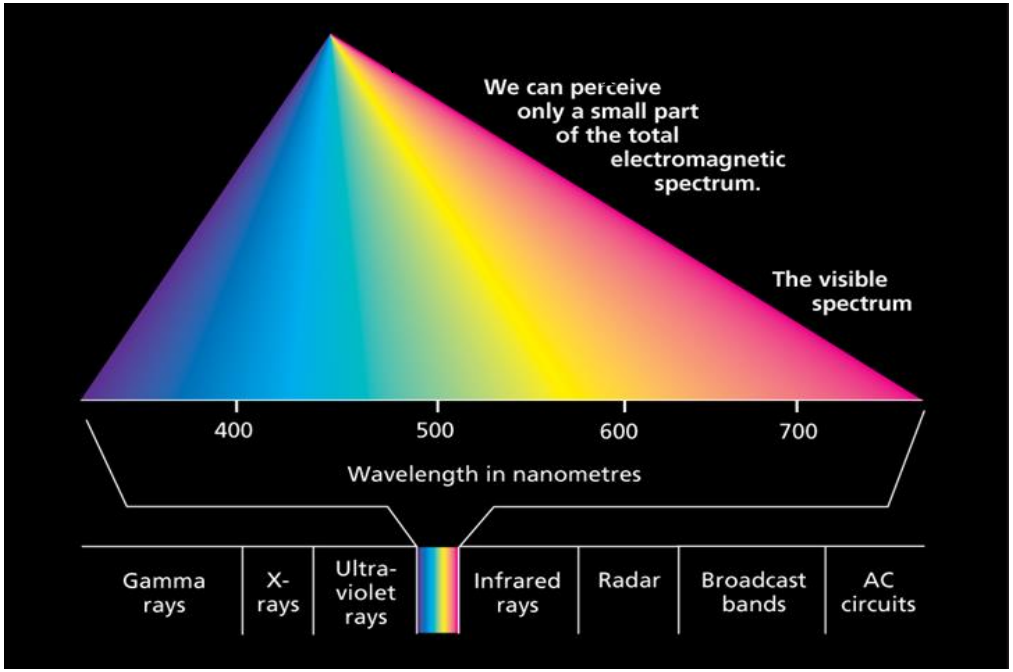


Vision

- What we see (Hue, Brightness, Saturation)
- An eye on the eye (Cornea, lens, Iris, Pupil, Retina, Rods & Cones)
- How we see colors (Trichromatic theory, Opponent process theory)

Psychological Properties of Light

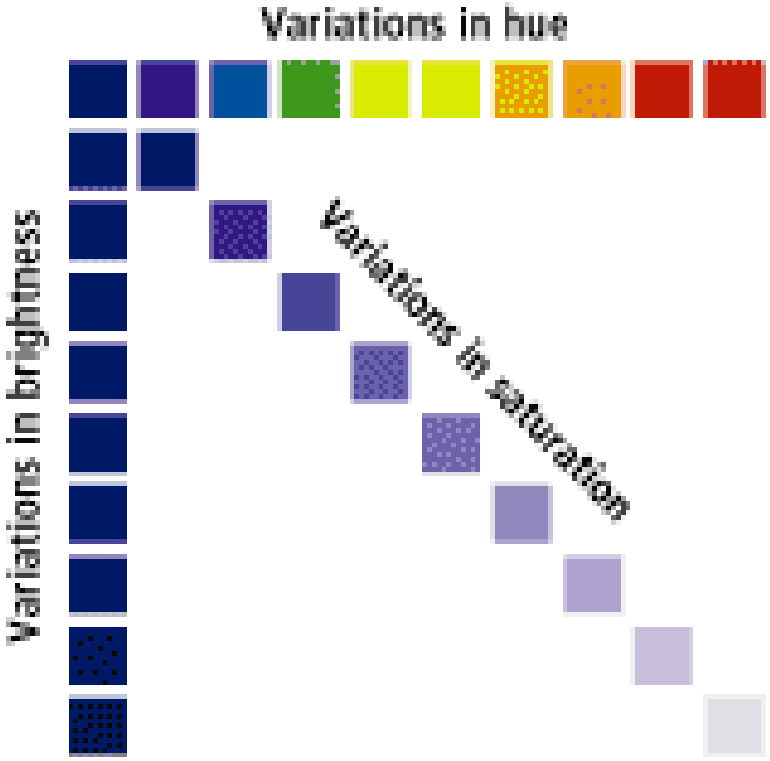
- Light – particles (Issac Newton) or Waves (Thomas Young)
- Light = Both particle + Waves = PHOTONS (Albert Einstein)



The wavelengths that people can see are only a small part of the whole electromagnetic spectrum.

What We See

- Hue
- Brightness
- Saturation

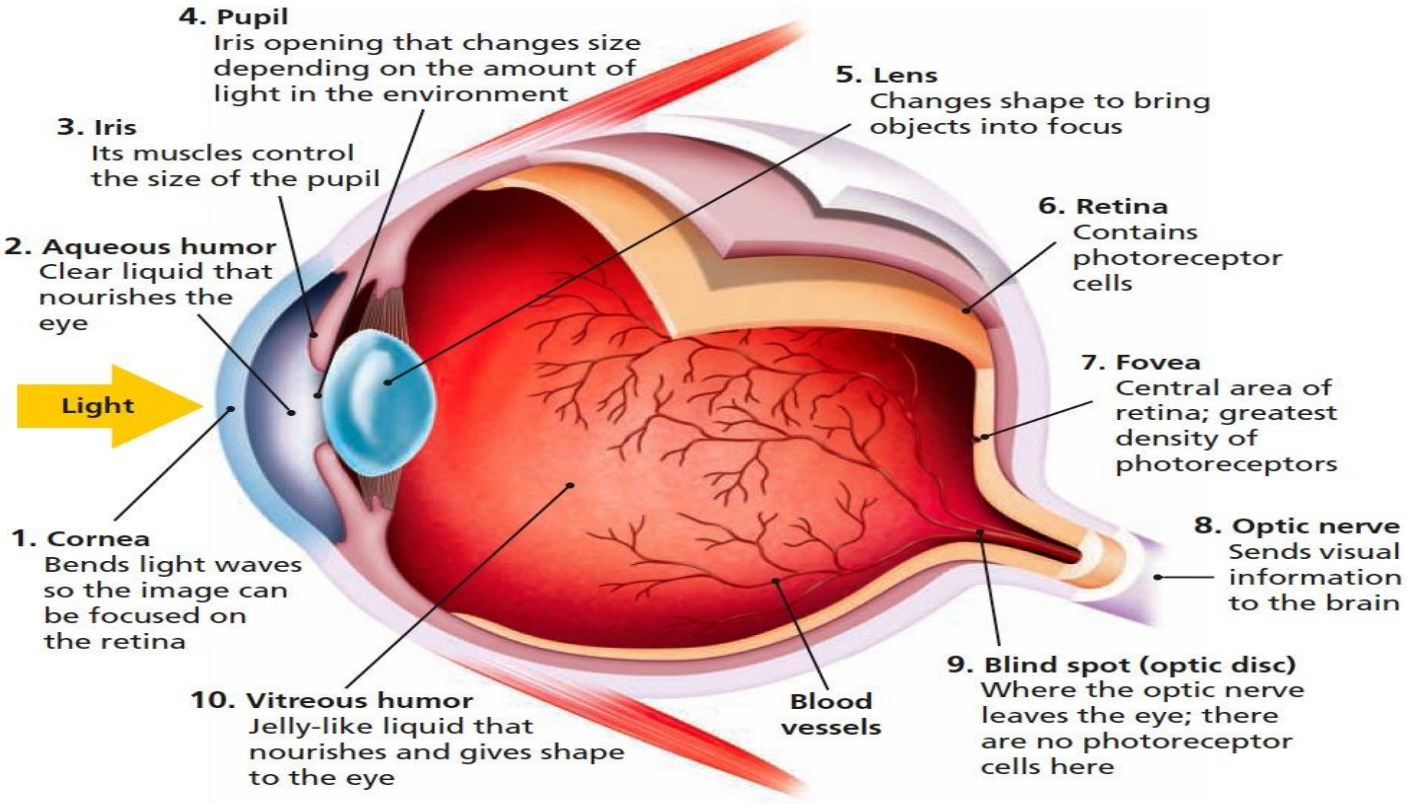


What We See

- Hue
  - Visual experience specified by colour names and related to the wavelength of light. Determined by length of the wave
- Brightness
  - Lightness and luminance; the visual experience related to the amount of light emitted from or reflected by an object. Determined by amplitude (high or low) of the wave. Higher the wave brighter the light.
- Saturation
  - Vividness or purity of colour; the visual experience related to the complexity of light waves. Determined by whether or how much mix of wavelengths is present.

An Eye on the Eye

Light enters the eye through the cornea and pupil. The iris controls the size of the pupil. From the pupil, light passes through the lens to the retina, where it is transformed into nerve impulses. The nerve impulses travel to the brain along the optic nerve.

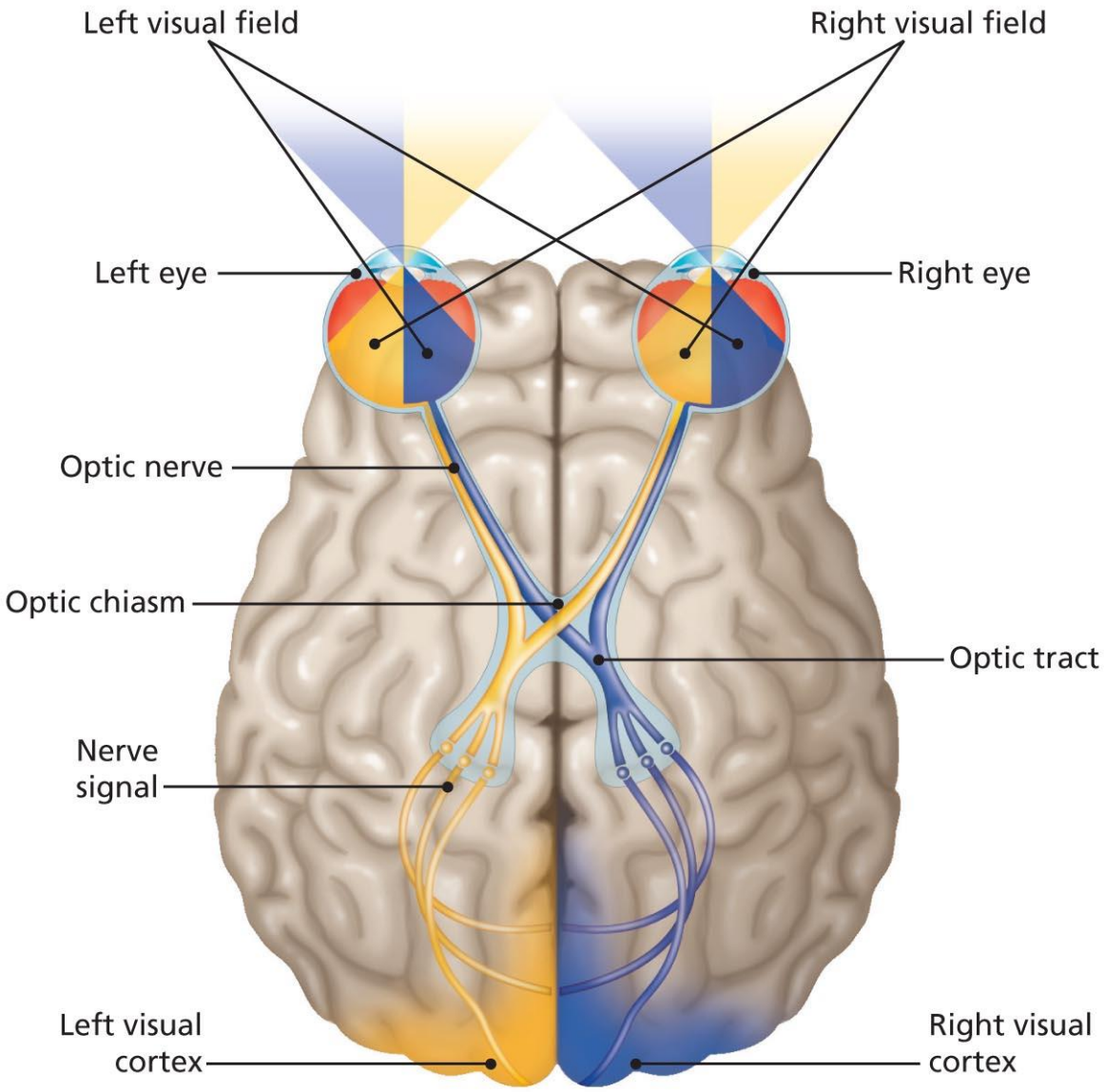


Structure of the eye

- Cornea
  - Protects eye and bends light toward lens.
- Lens
  - Focuses on objects by changing shape.
- Iris
  - Controls amount of light that gets into eye.
- Pupil
  - Widens or dilates to let in more light.
- Retina
  - Neural tissue lining the back of the eyeball's interior, which contains the receptors for vision.
- Rods
  - Visual receptors that respond to dim light.
- Cones
  - Visual receptors involved in colour vision. Most humans have 3 types of cones.

Crossing of the Optic Nerve

Light falling on the left side of each eye's retina (from the right visual field, shown in yellow) will stimulate a neural message that will travel along the optic nerve to the thalamus, and then on to the visual cortex in the occipital lobe of the left hemisphere. Notice that the message from the temporal half of the left retina goes to the left occipital lobe, while the message from the nasal half of the right retina crosses over to the left hemisphere (the optic chiasm is the point of crossover). The optic nerve tissue from both eyes joins together to form the left optic tract before going on to the thalamus and the left occipital lobe. For the left visual field (shown in blue), the messages from both right sides of the retinas will travel along the right optic tract to the right visual cortex in the same manner.



Why the Visual System is not a Camera

- Much visual processing is done in the brain.
  - Some cortical cells respond to lines in specific orientations (e.g. horizontal).
  - Other cells in the cortex respond to other shapes (e.g., bulls-eyes, spirals, faces).

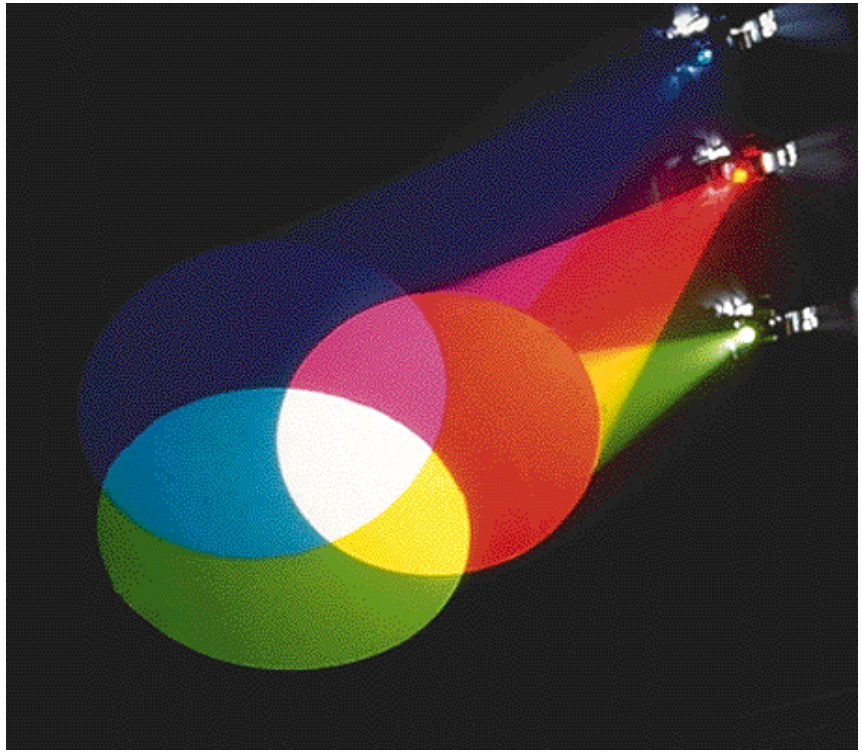
- Feature-detectors
  - Cells in the visual cortex that are sensitive to specific features of the environment.

How We See Colours

- Trichromatic theory
- Opponent process theory

Trichromatic Theory

- Young (1802) & von Helmholtz (1852) both proposed that the eye detects 3 primary colours:
  - red, blue, & green
- All other colours can be derived by combining these three.



Opponent-Process Theory

- First proposed by German physiologist Ewald Hering in the 19th century
- A competing theory of colour vision, which assumes that the visual system treats pairs of colours as opposing or antagonistic.
- Opponent-Process cells are inhibited by a colour, and have a burst of activity when it is removed.

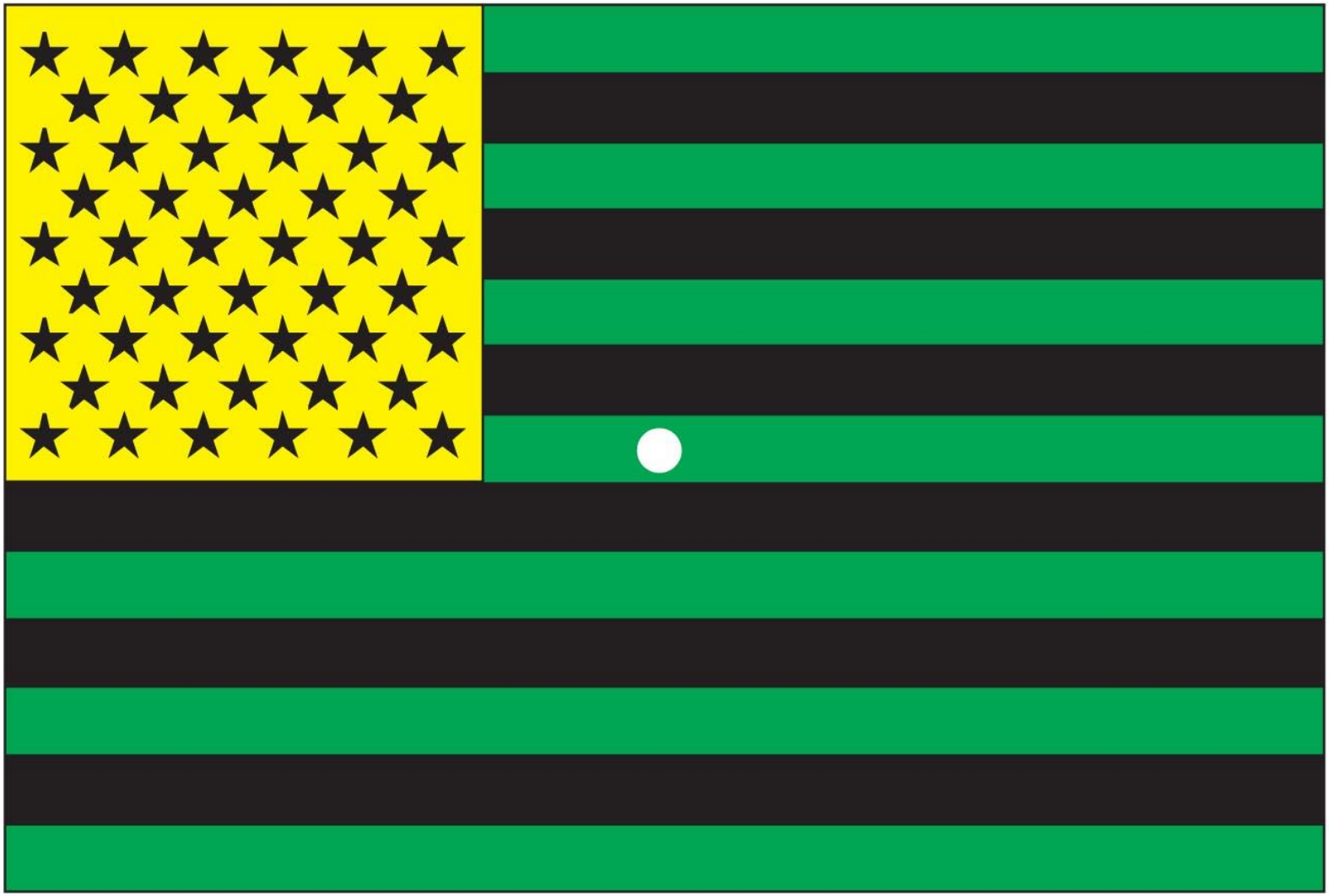
Color Vision

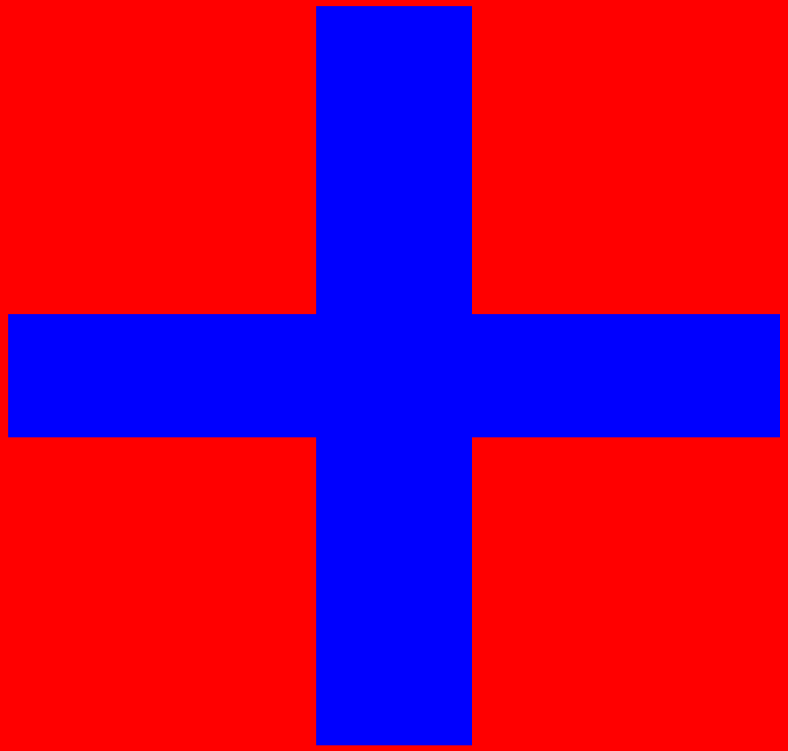
- Evidence for opponent-process theory:

afterimages: images that occur when a visual sensation persists for a brief time even after the original stimulus is removed

Color Afterimage

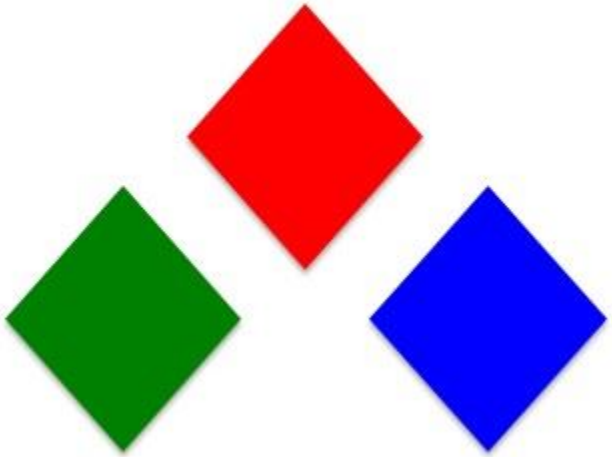
Stare at the white dot in the center of this oddly colored flag for about 30 seconds. Now look at a white piece of paper or a white wall. Notice that the colors are now the normal, expected colors of the American flag. They are also the primary colors that are opposites of the colors in the picture and provide evidence for the opponent-process theory of color vision.



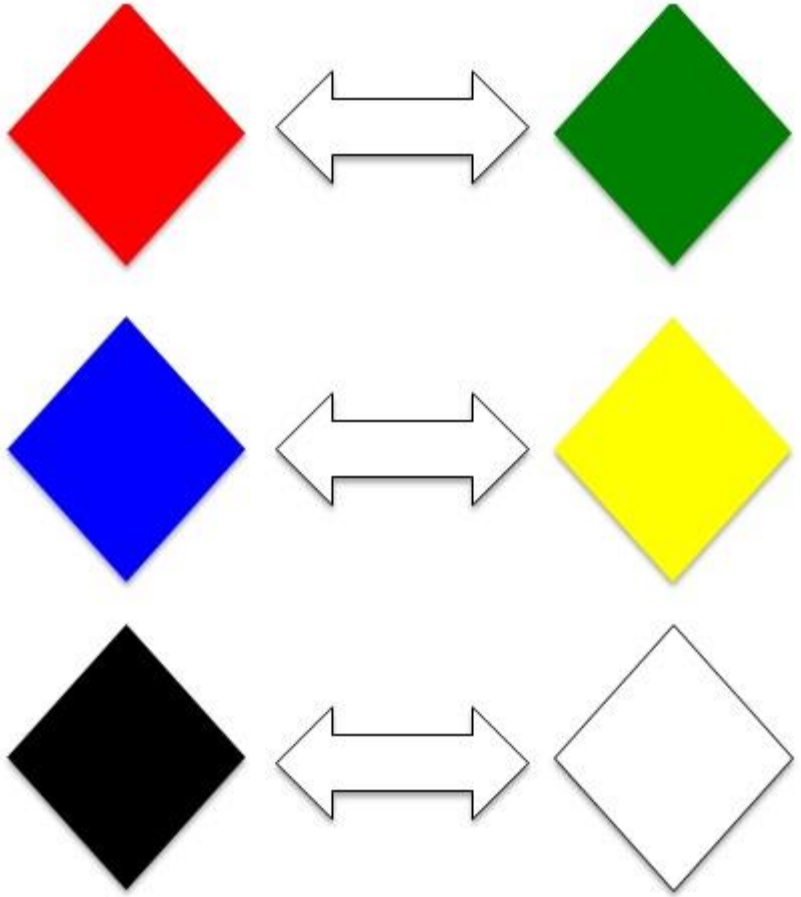


ColoUr Vision

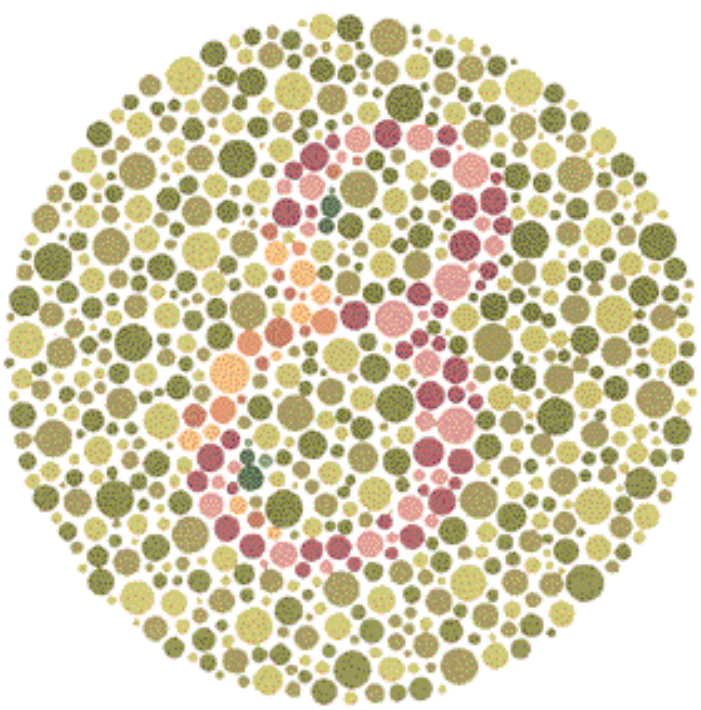
Trichromatic Theory

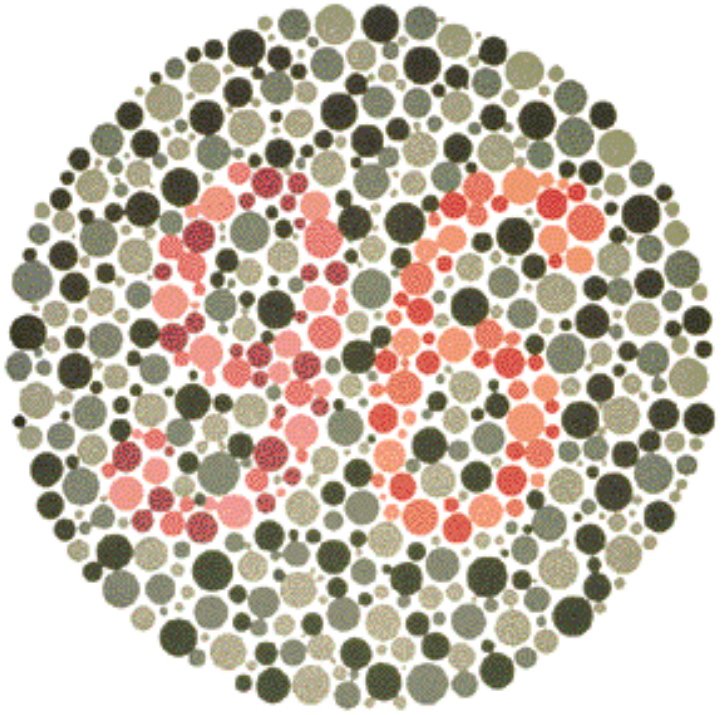


Opponent-Process Theory



Test of Colour Deficiency





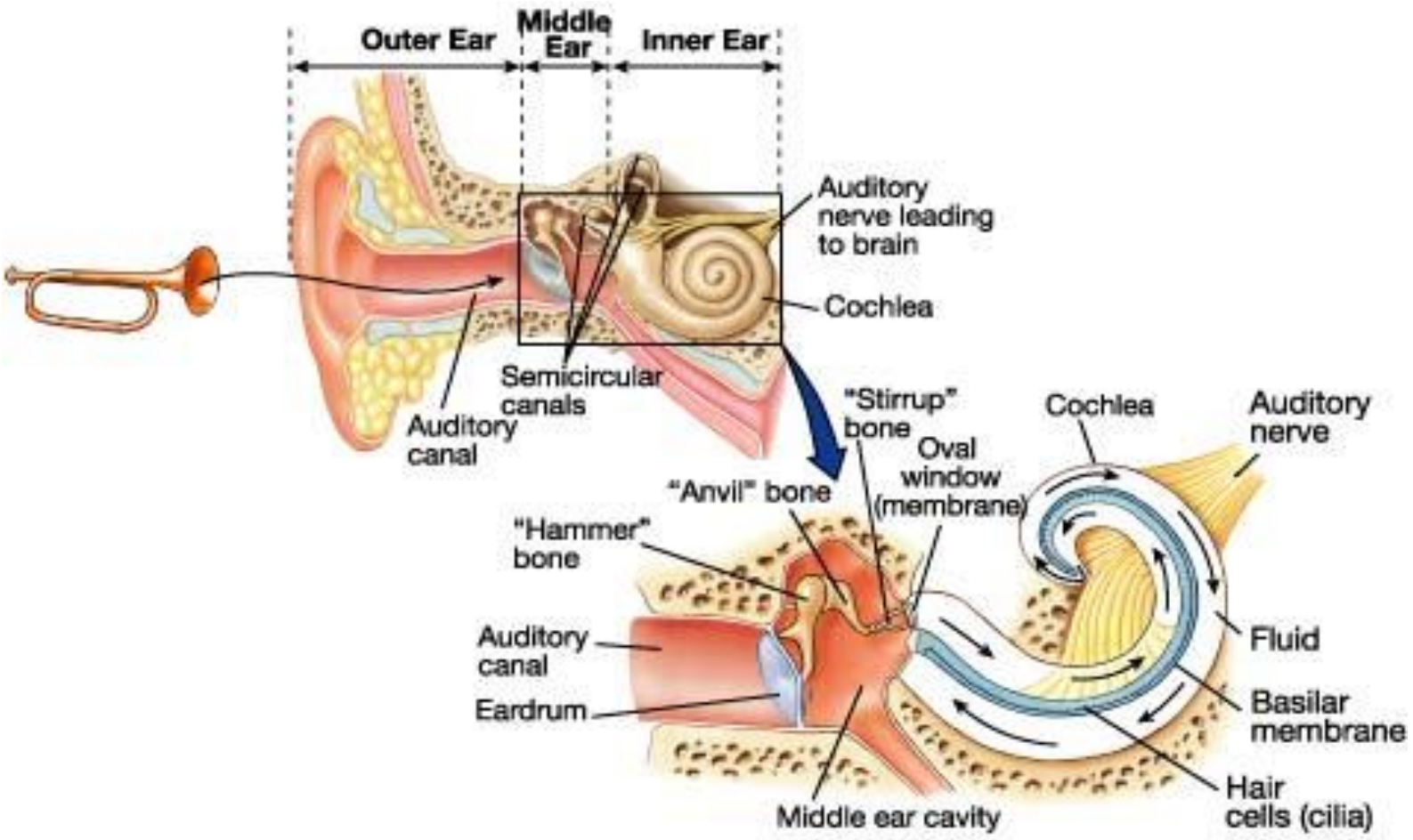
Hearing

- What we hear
- An ear on the world
- Constructing the auditory world

What We Hear:
The Physical Aspects of Sound

- Loudness
  - The dimension of auditory experience related to the intensity of a pressure wave.
- Pitch
  - The dimension of auditory experience related to the frequency of a pressure wave.
- Timbre (pronounced "TAM-bur")
  - The distinguishing quality of sound; the dimension of auditory experience related to the complexity of the pressure wave.

An Ear on the World



Sensing Sound

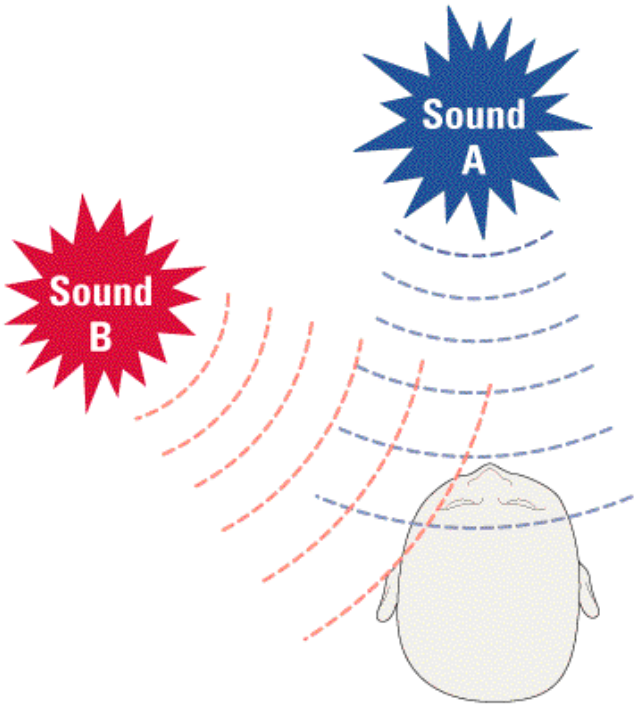
- **Sound**
  - Movement of air molecules brought about by a source of vibration
- **Eardrum**
  - Vibrates when sound waves hit it
  - **Middle ear -** Hammer, anvil, stirrup

Sensing Sound

- **Inner Ear**
  - Changes sound vibrations into a form in which they can be transmitted to the brain
    - **Cochlea**
      - Filled with fluid and vibrates in response to sound
    - **Basilar membrane**
      - Dividing cochlea into an upper chamber and lower chamber
      - Covered with *hair cells*

Auditory Localization

- Sounds from different directions are not identical as they arrive at left and right ears
  - Loudness
  - Timing
  - Phase
- The brain calculates a sound's location by using these differences.



Sorting Out Theories of Sound

- **Place Theory of Hearing**
  - States that different areas of the basilar membrane respond to different frequencies

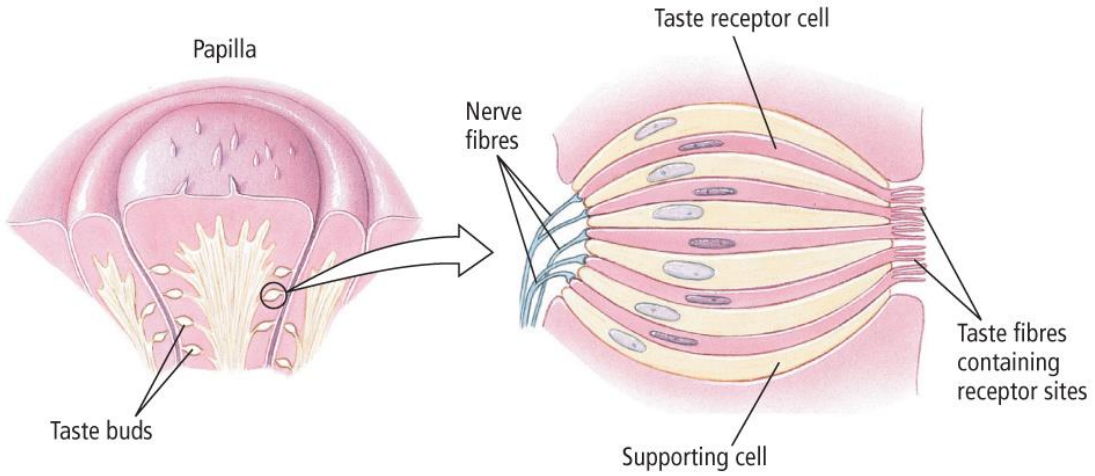
- **Frequency Theory of Hearing**
  - Suggests that the entire basilar membrane acts like a microphone, vibrating as a whole in response to a sound

Other Senses

- Taste: savoury sensations
- Smell: The sense of scents
- Senses of the skin
- The mystery of pain
- The environment within

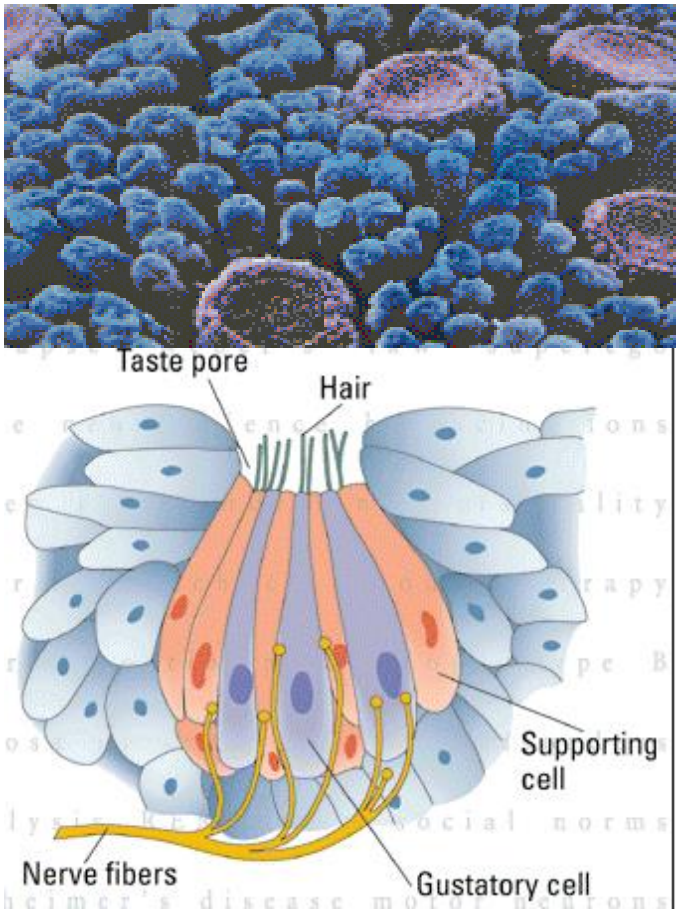
Taste: Savoury Sensations

- Papillae
  - Knoblike elevations on the tongue, containing the taste buds (Singular: papilla).
- Taste buds
  - Nests of taste-receptor cells.



Taste Buds

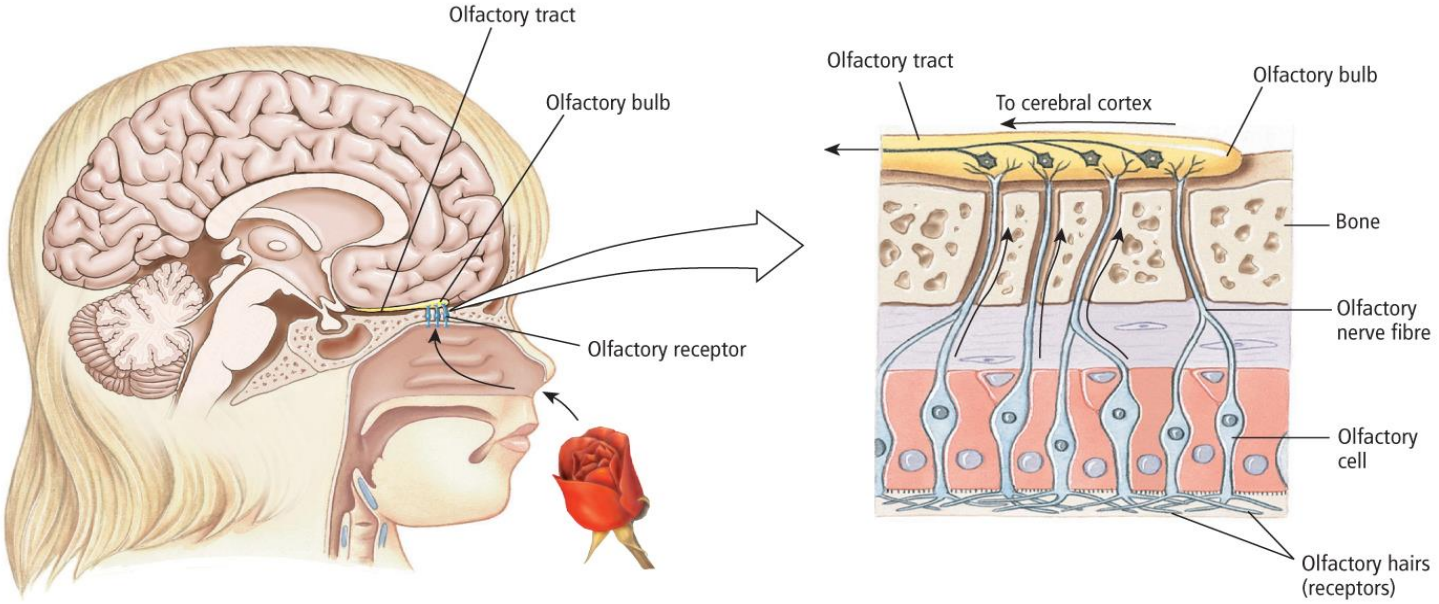
- Photograph of tongue surface (top), magnified 75 times.
- 10,000 taste buds line the tongue and mouth.
  - Taste receptors are down inside the "bud"
- Children have more taste buds than adults.



Four Tastes

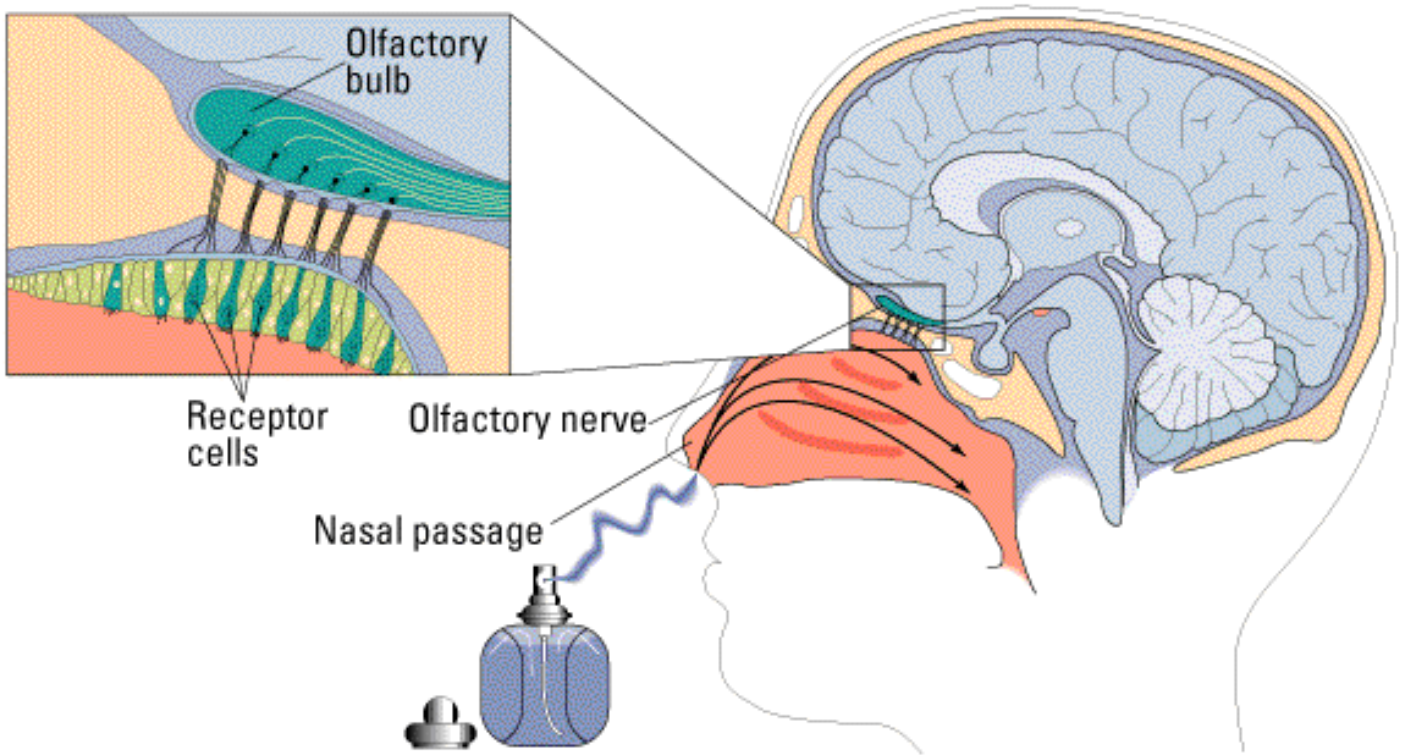
- Four basic tastes
  - Salty, sour, bitter and sweet.
- Different people have different tastes based on:
  - Genetics
  - Culture
  - Learning
  - Food attractiveness

Smell: The Sense of Scents



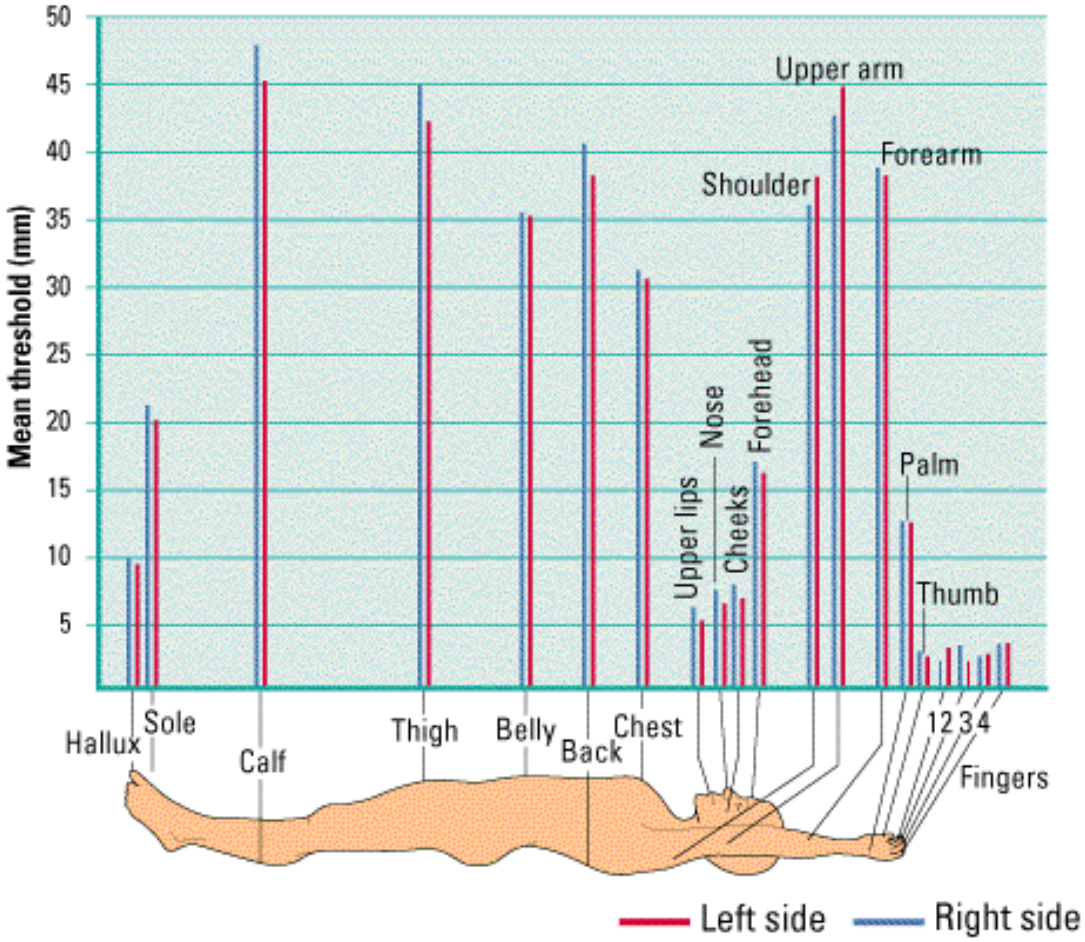
- Airborne chemical molecules enter the nose and circulate through the nasal cavity.
  - Vapors can also enter through the mouth and pass into nasal cavity.
- Receptors on the roof of the nasal cavity detect these molecules.

Olfactory System



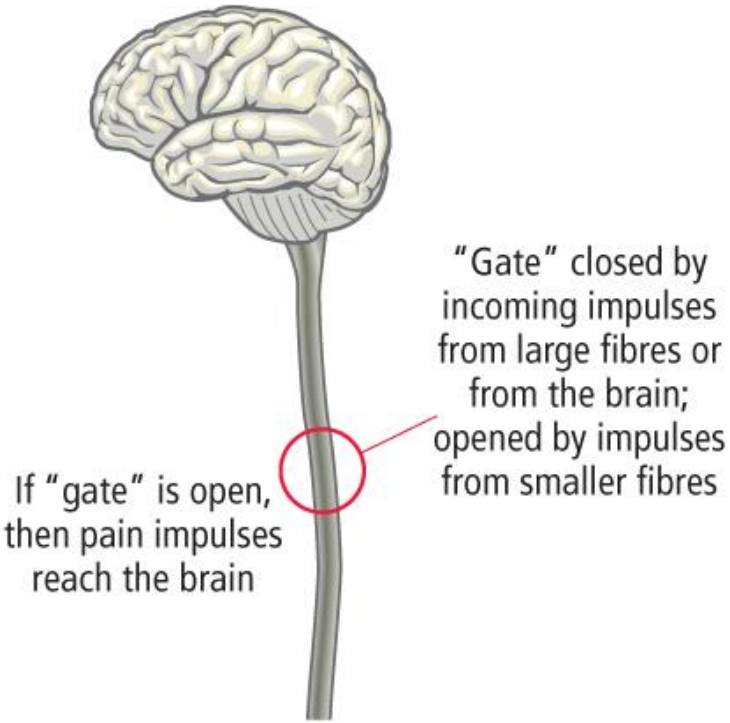
Sensitivity to Touch:

Touch, Pressure, Temperature and Pain



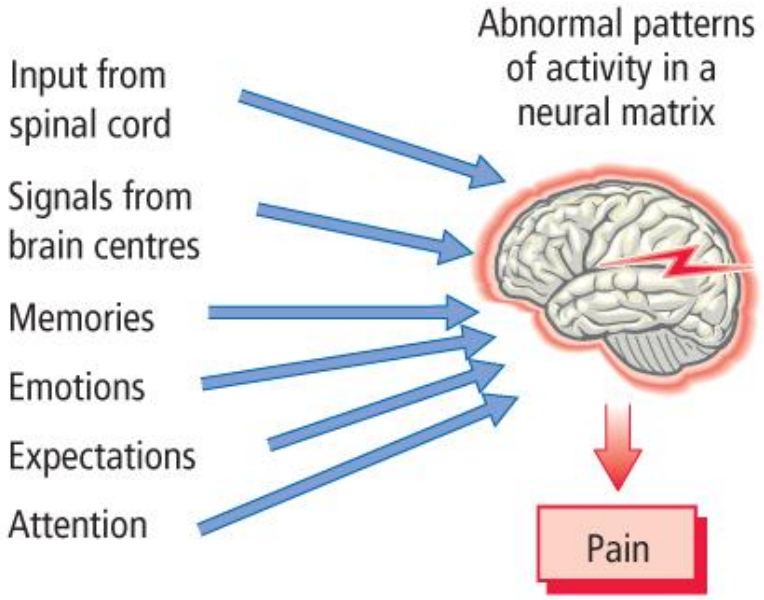
Gate-Control Theory of Pain

- Experience of pain depends (in part) on whether the pain impulse gets past neurological "gate" in the spinal cord and thus reaches the brain.
- Particular nerve receptors lead to specific areas of the brain related to pain.



Neuromatrix Theory of Pain

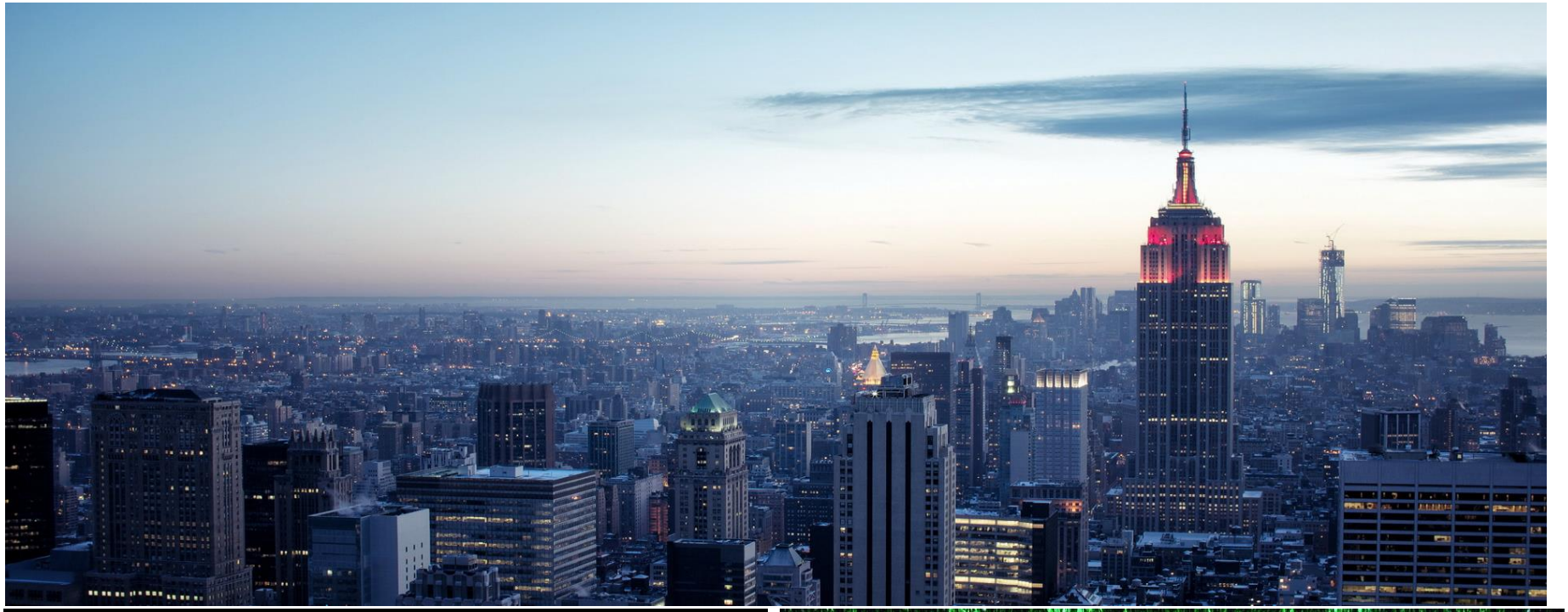
- Theory that the matrix of neurons in the brain is capable of generating pain (and other sensations) in the absence of signals from sensory nerves.

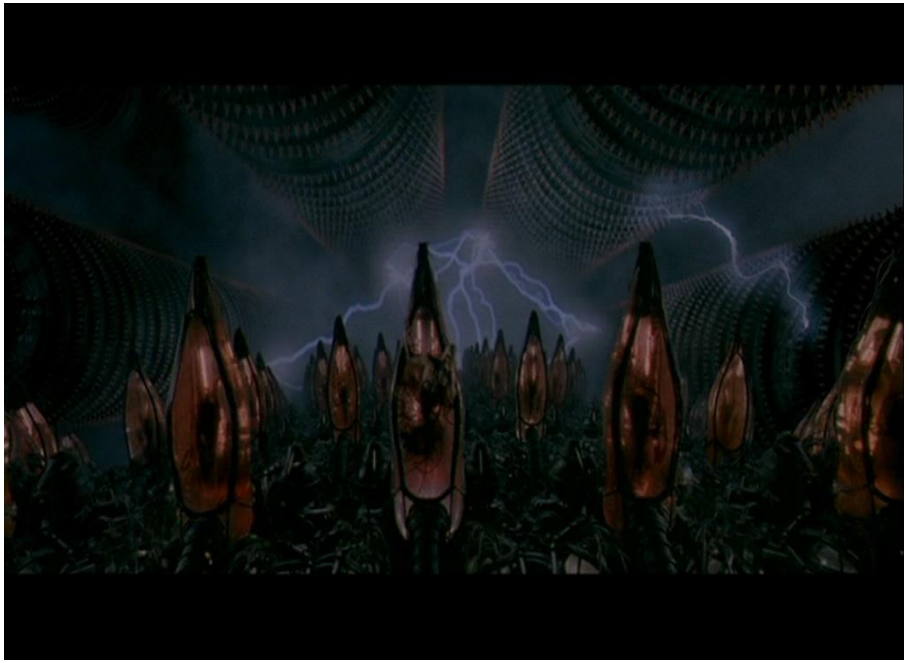


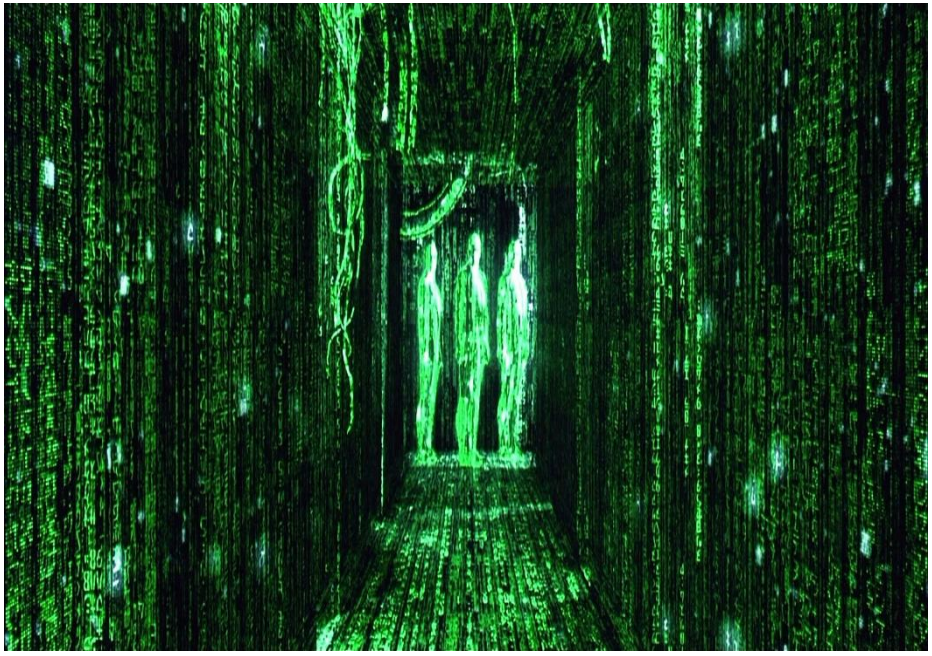
The Environment Within

- Kinesthesis
  - The sense of body position and movement of body parts; also called kinesthesia.
- Equilibrium
  - The sense of balance.
- Semicircular Canals
  - Sense organs in the inner ear, which contribute to equilibrium by responding to rotation of the head.

Perception







Key Questions

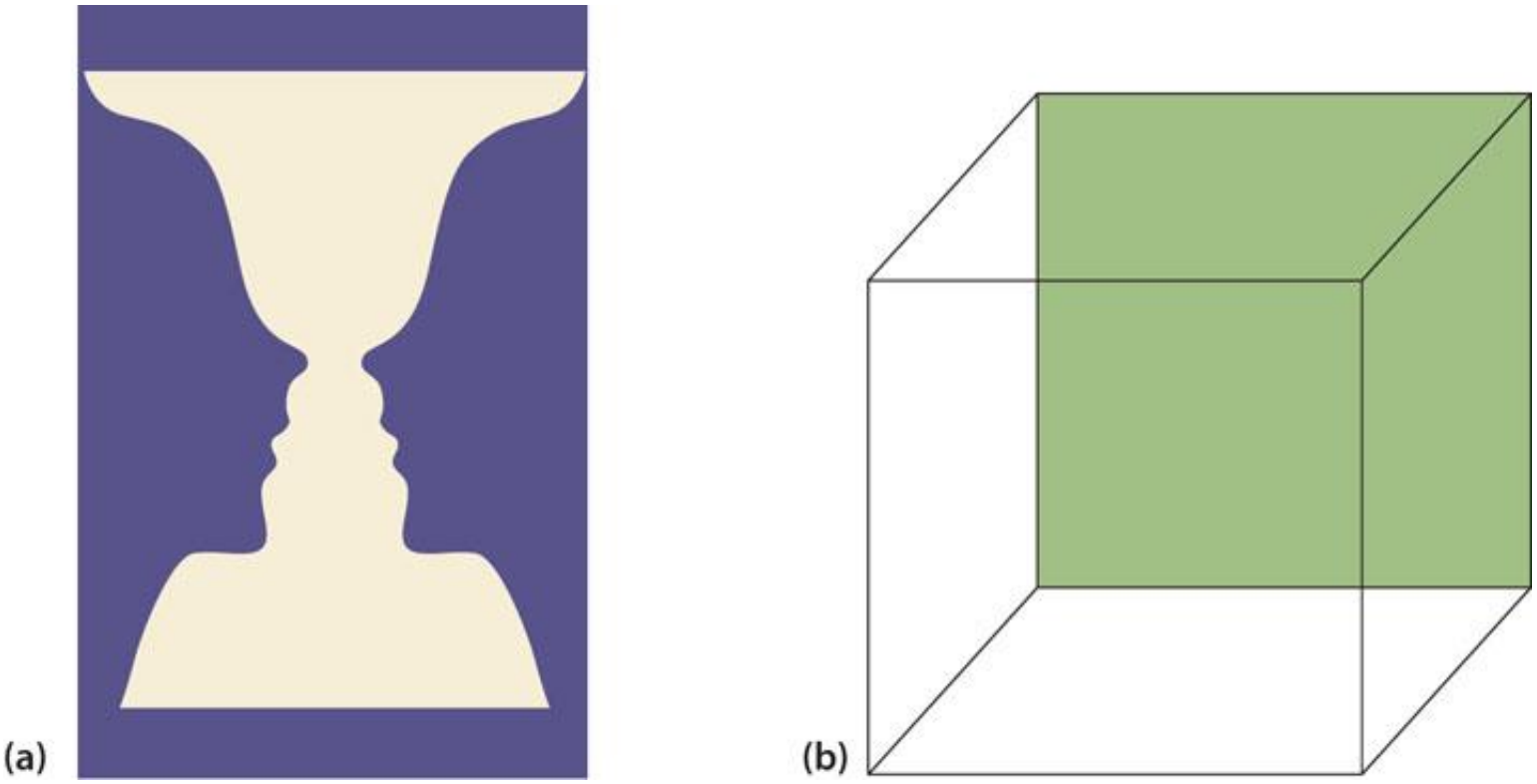
- What principles underlie our organization of the visual world and allow us to make sense of our environment?

- How are we able to perceive the world in three dimensions when our retinas are capable of sensing only two dimensional images?

- What clues do visual illusions give us about our understanding of general perceptual mechanisms?

The Gestalt Laws of Organization

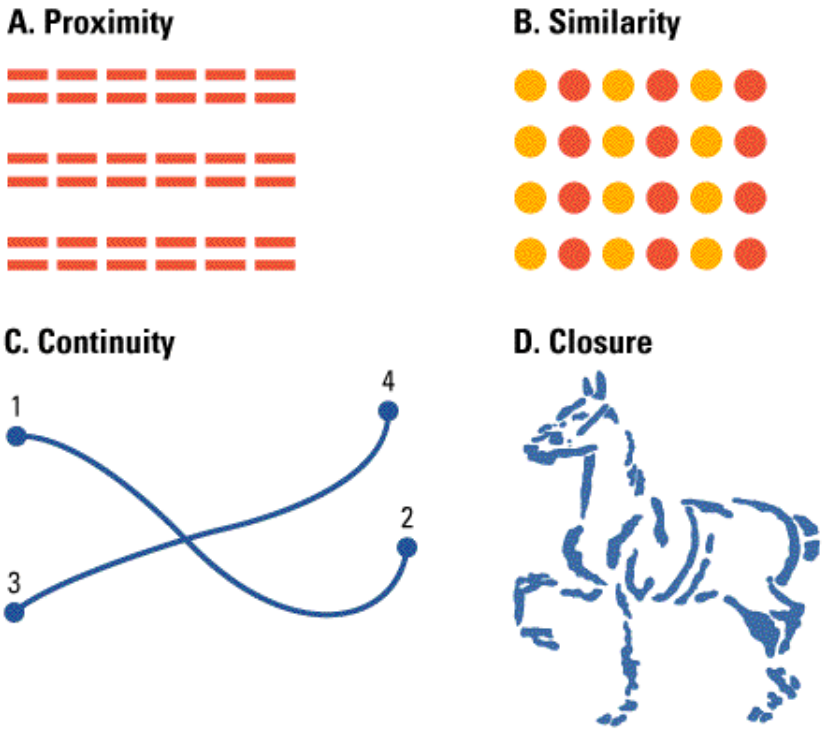
- Series of principles that focus on the ways we organize bits and pieces of information into meaningful wholes
  - *Gestalt (a German word for form or shape)*



When the usual cues we use to distinguish figure from ground are absent, we may shift back and forth between different views of the same figure. If you look at each of these objects long enough, you'll probably experience a shift in what you're seeing. In (a), you can see either a vase or the profiles of two people. In (b), the shaded portion of the figure, called a Necker cube, can appear to be either the front or the back of the cube.

The Gestalt Laws of Organization

- Proximity
  - Seeing 3 pair of lines in A.
- Similarity
  - Seeing columns of orange and red dots in B.
- Continuity
  - Seeing lines that connect 1 to 2 and 3 to 4 in C.
- Closure
  - Seeing a horse in D.



The Gestalt Laws of Organization-explained



- *a. closure*: We usually group elements to form enclosed or complete figures rather than open ones.
- *b. proximity*: We perceive elements that are closer together as grouped together.
- c. Elements that are *similar* in appearance we perceive as grouped together.
- *d. simplicity*: When we observe a pattern, we perceive it in the most basic, straightforward manner that we can.

Try Reading this sentence.

Ca- yo- re-d t-is -en-en-e, w-ic- ha- ev-ry -hi-d l-tt-r m-ss-ng?

Perception is an active, constructive process.

Sensation ⟷ **Perception**



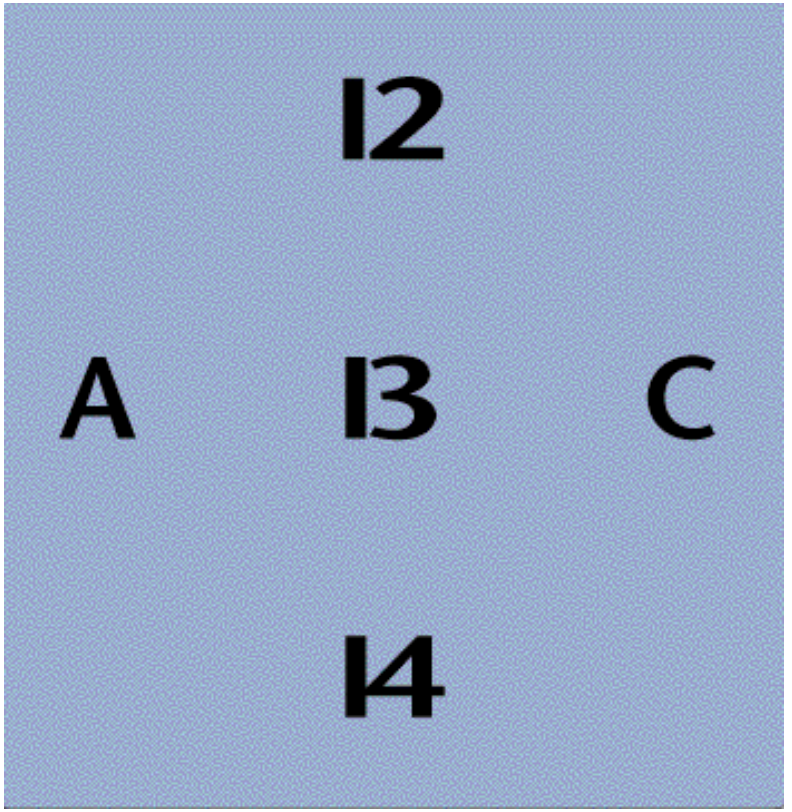
Top-Down and Bottom-Up Processing

- Top-Down Processing
  - Perception is guided by higher-level knowledge, experience, expectations, and motivations

- Bottom-Up Processing
  - Consists of the progression of recognizing and processing information from individual components of a stimulus and moving to the perception of the whole

Context Effects

- The same physical stimulus can be interpreted differently
- We use other cues in the situation to resolve ambiguities
- Is this the letter B or the number 13?



The power of context



Top-down and bottom-up processing occur simultaneously, and interact with each other, in our perception of the world around us. Bottom-up processing permits us to process the fundamental characteristics of stimuli, whereas top-down processing allows us to bring our experience to bear on perception.

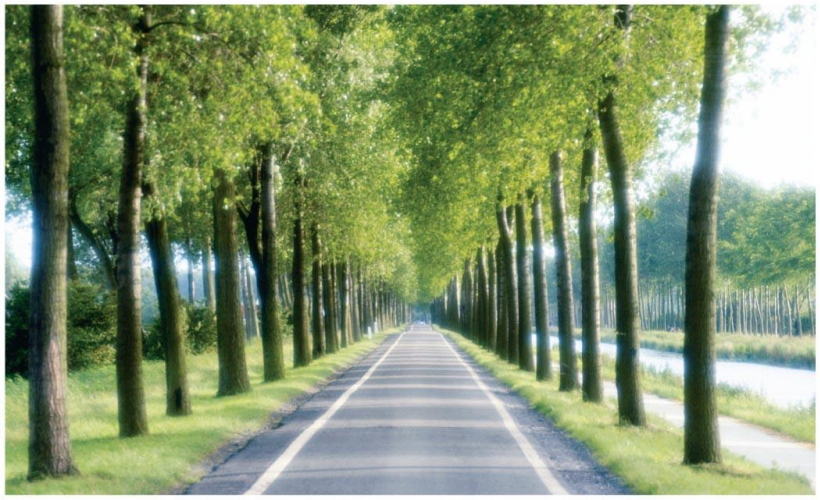
Different Process of perception

Depth Perception: Translating 2D into 3D

- Ability to view the world in three dimensions and to perceive distance
  - Largely due to the fact that we have two eyes
    - Monocular cues
      - Motion parallax
      - Relative size
      - Texture gradient
      - Linear perspective
    - Binocular cues

Monocular Cues

- Monocular cues (pictorial depth cues): cues for perceiving depth based on one eye only



a.

linear perspective: the tendency for parallel lines to appear to converge on each other



d.

relative size: perception that occurs when objects that a person expects to be of a certain size appear to be small and are, therefore, assumed to be much farther away



c.

aerial (atmospheric) perspective: the haziness that surrounds objects that are farther away from the viewer, causing the distance to be perceived as greater

texture gradient: the tendency for textured surfaces to appear to become smaller and finer as distance from the viewer increases



b.

Monocular Cues

- Overlap (or Interposition): the assumption that an object that appears to be blocking part of another object is in front of the second object and closer to the viewer



Monocular Cues

When you are riding in a car for example, the nearby telephone poles rush by much faster than the trees or buildings in the distance.



- motion parallax: the perception of motion of objects in which close objects appear to move more quickly than objects that are farther away.

Monocular Cues

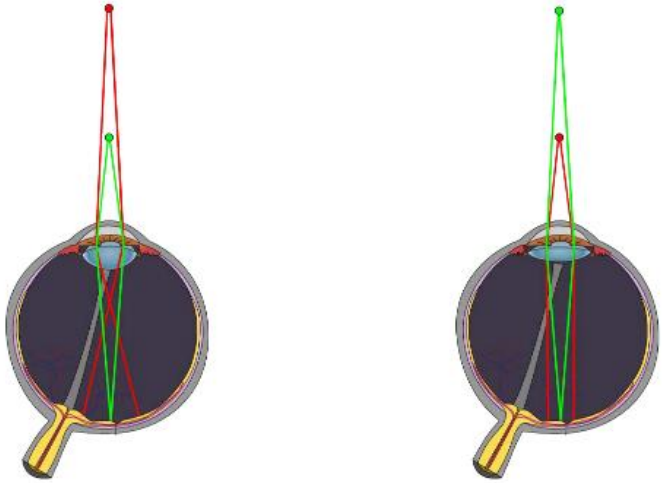


Figure A.2: Right eye accommodation on a) near and b) far target.

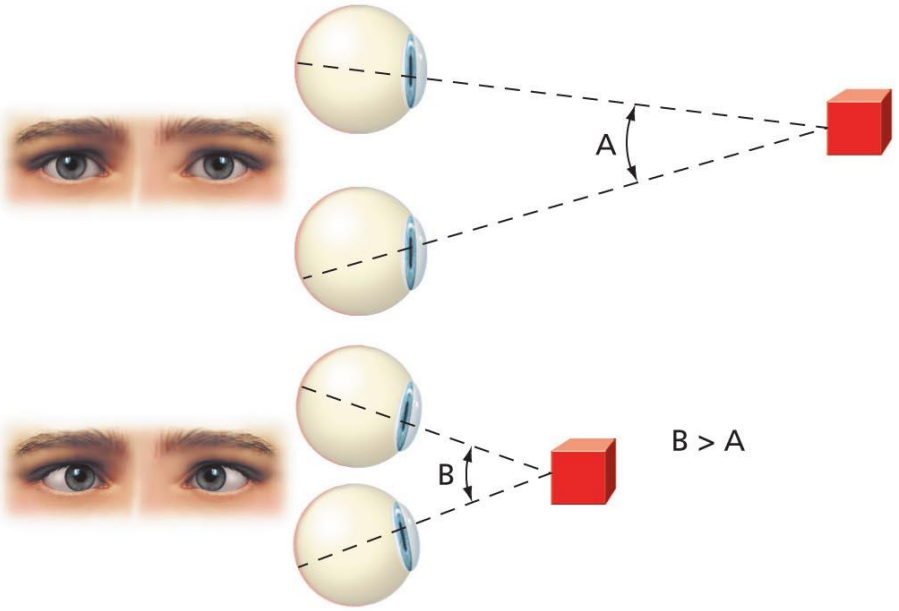
- accommodation: as a monocular clue, the brain's use of information about the *changing thickness of the lens of the eye* in response to looking at objects that are close or far away

Binocular Cues

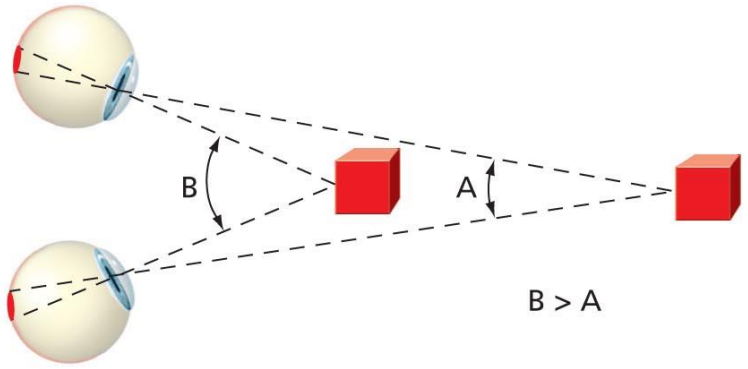
- Binocular cues: cues for perceiving depth based on both eyes

Binocular Cues

Binocular Cues to Depth Perception



Convergence



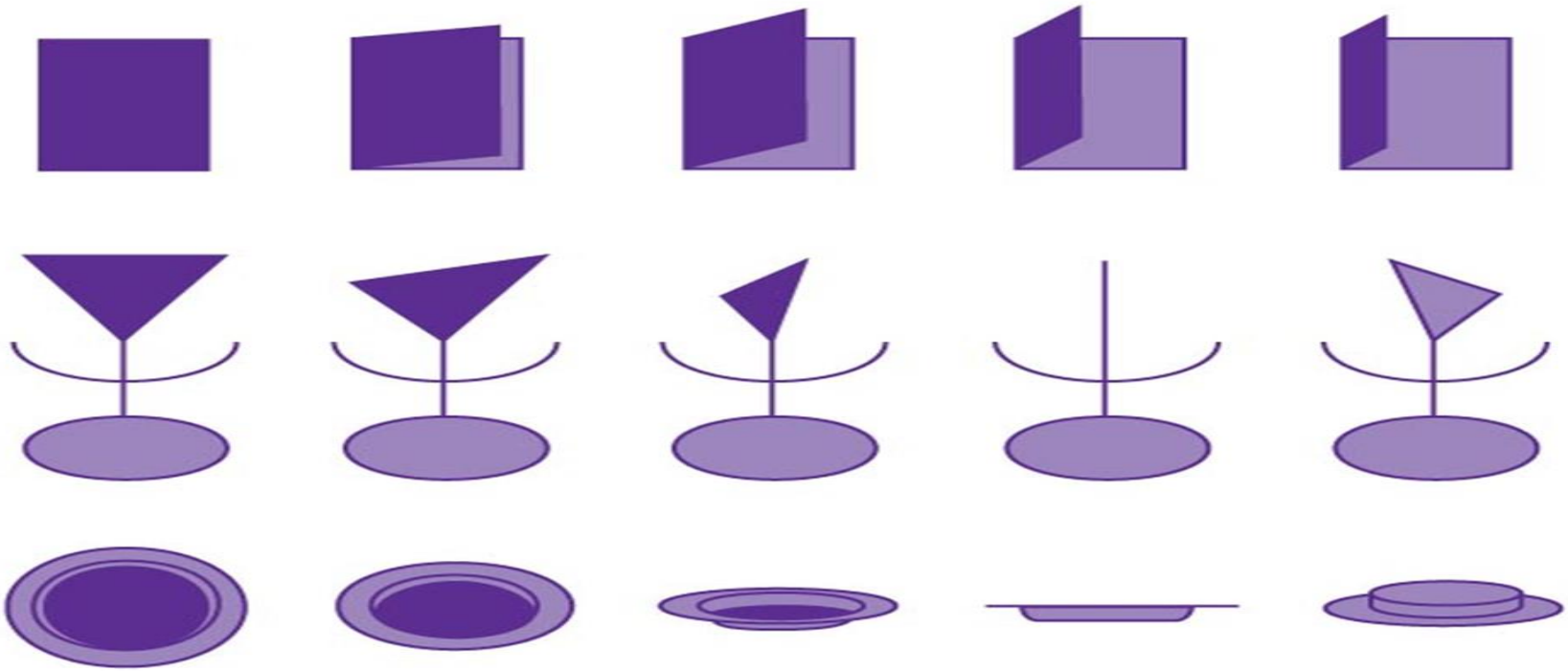
Binocular disparity

convergence: the rotation of the two eyes in their sockets to focus on a single object, resulting in greater convergence for closer objects and lesser convergence if objects are distant

binocular disparity: the difference in images between the two eyes, which is greater for objects that are close and smaller for distant objects

Perception of constancy and motion

Perceptual Constancy



- Phenomenon in which physical objects are perceived as unvarying and consistent despite changes in their appearance or in the physical environment

Motion Perception:
As the World Turns

- **Cues about perception of motion**
  - The movement of an object across the retina is typically perceived relative to some stable, unmoving background
  - Movement of images across the retina
  - We factor in information about our own head and eye movements, along with information about changes in the retinal image

Motion Perception:
As the World Turns

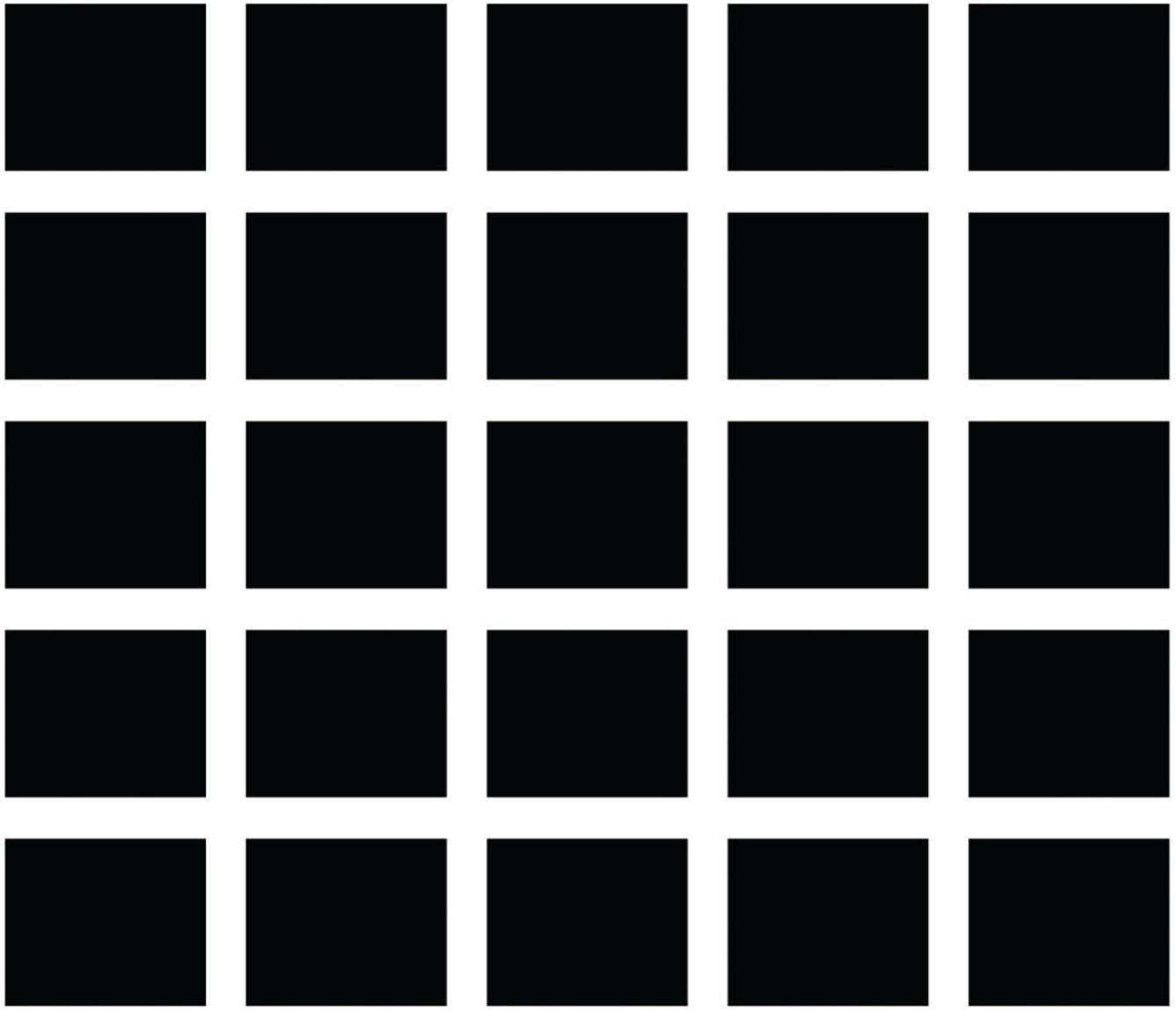
- ***Apparent movement –***

  The perception that a stationary object is moving. It occurs when different areas of the retina are quickly stimulated, leading us to interpret motion.

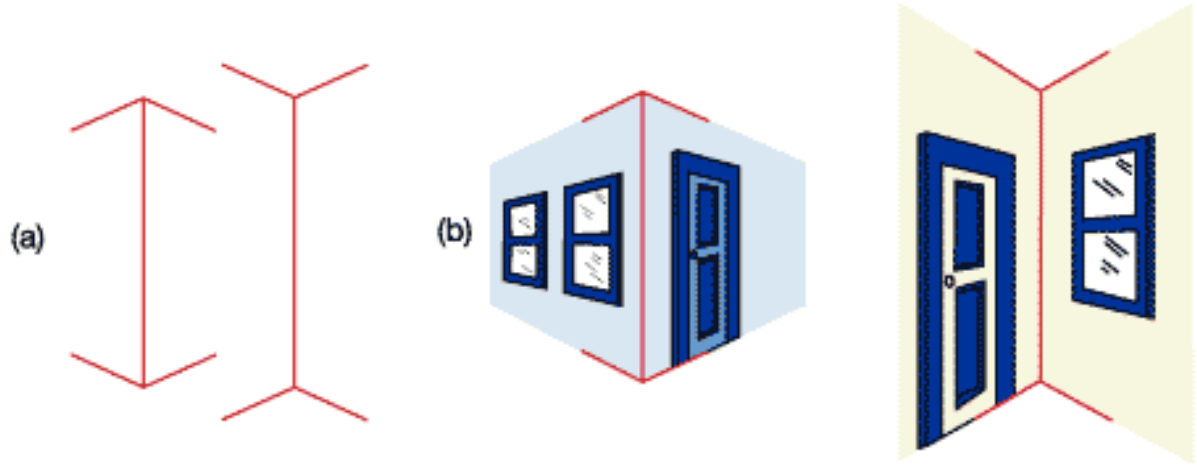
Perceptual Illusions

- Illusions are valuable in understanding perception because they are systematic errors.
  - Illusions provide hints about perceptual strategies.

The Hermann Grid



possibly due to the response of the primary visual cortex



- In the Muller-Lyer illusion (above) we tend to perceive the line on the right as slightly longer than the one on the left.

Illusions of motion

- autokinetic effect: a small, stationary light in a darkened room will appear to move or drift because there are no surrounding cues to indicate that the light is not moving

- stroboscopic motion: seen in motion pictures, in which a rapid series of still pictures will appear to be in motion

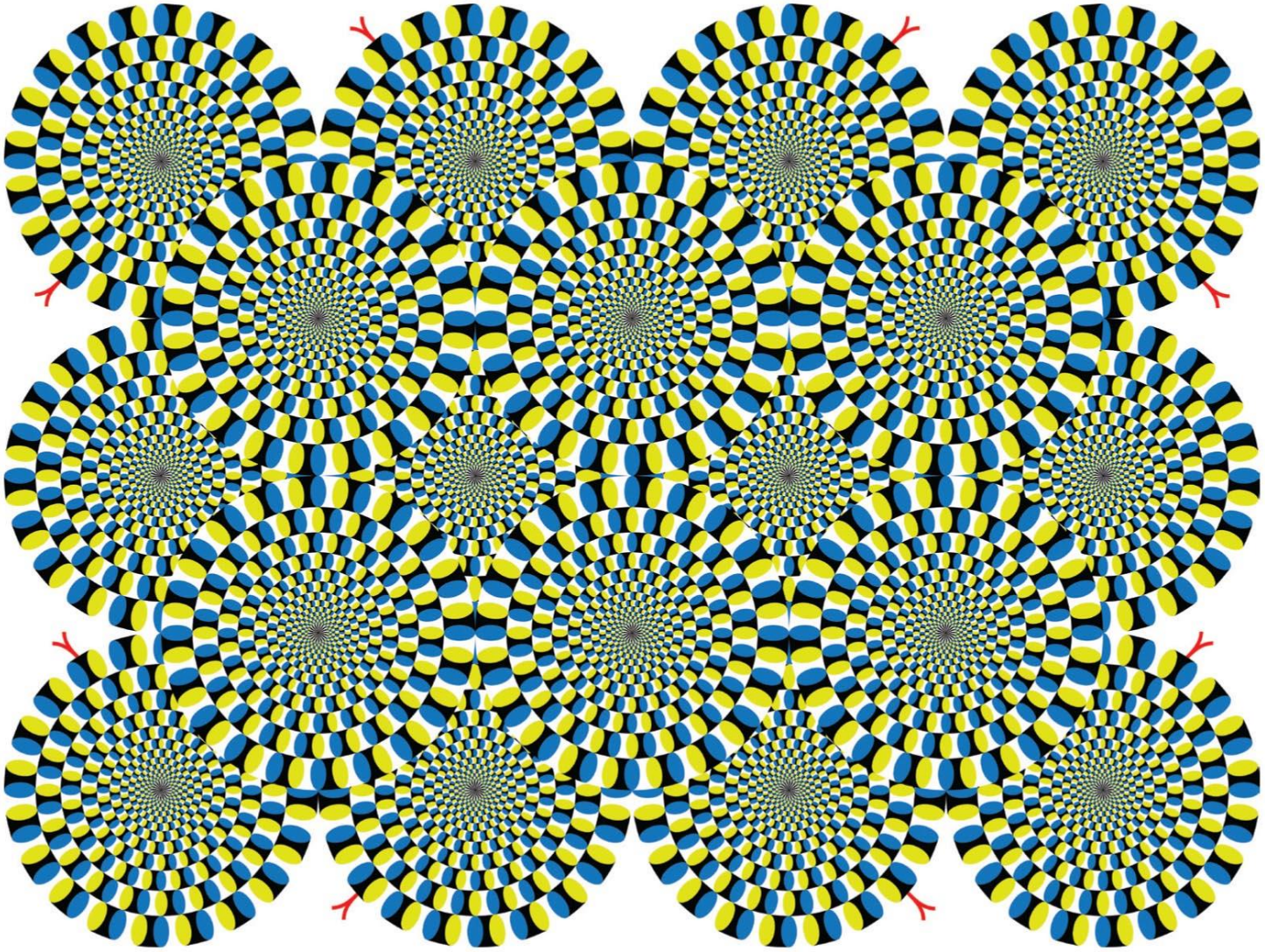
Illusions of motion

- Phi phenomenon: lights turned on in a sequence appear to move
- Rotating snakes: due in part to eye movements
- *The Enigma*: due in part to microsaccades

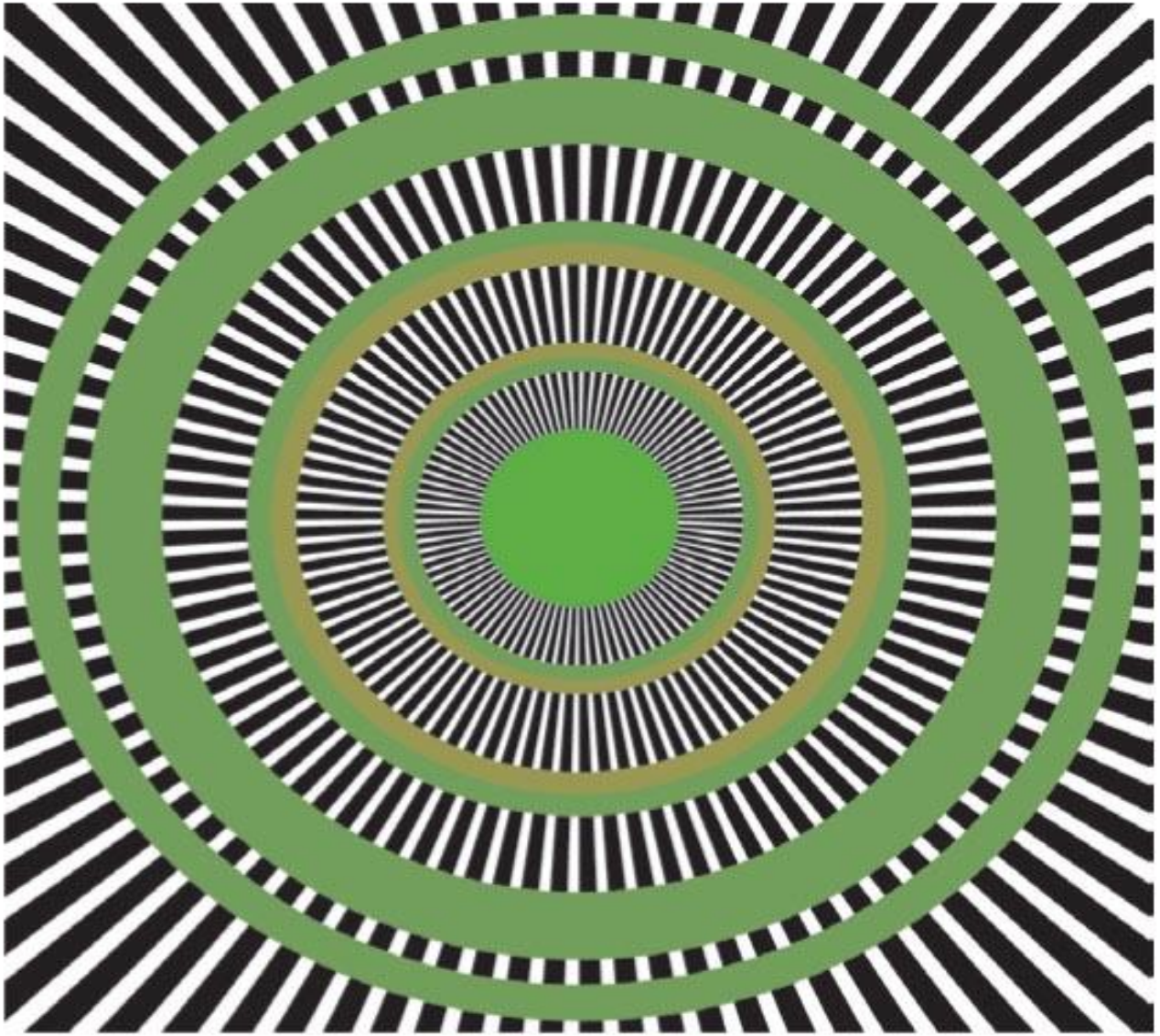
(i.e. a kind of fixational eye movement. They are small, jerk-like, involuntary eye movements, similar to miniature versions of voluntary saccades).



phi phenomenon: lights turned on in a sequence appear to move



rotating snakes: due in part to eye movements



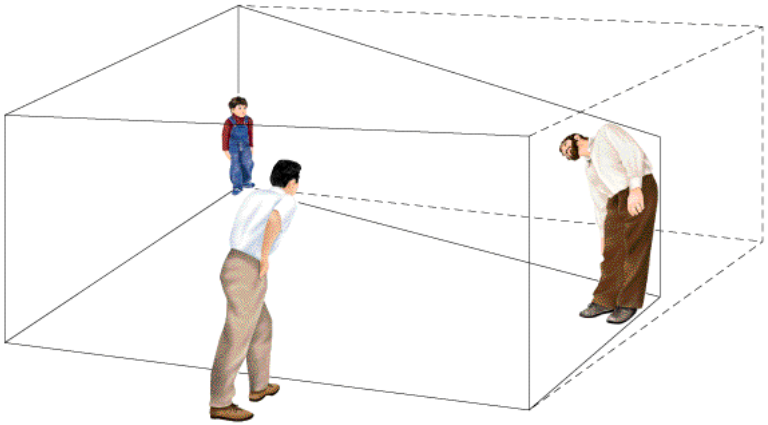
The Enigma: due in part to microsaccades (i.e. a kind of fixational eye movement. They are small, jerk-like, involuntary eye movements, similar to miniature versions of voluntary saccades).



- Ames Room Illusion

The Ames Room

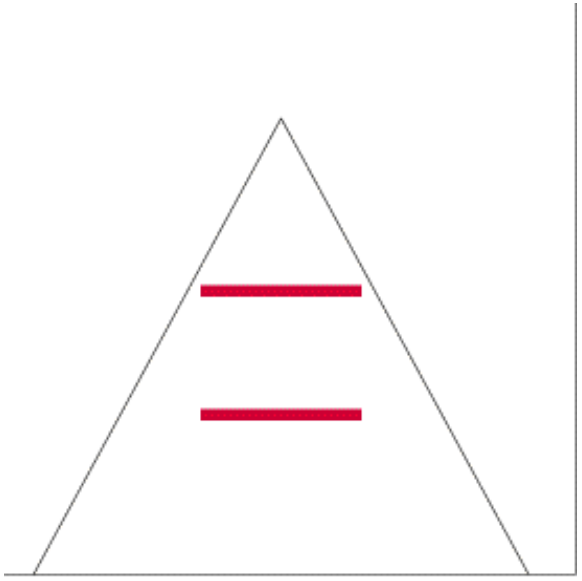
- A specially-built room that makes people seem to change size as they move around in it
- The room is not a rectangle, as viewers assume it is
- A single peephole prevents using binocular depth cues



The Ponzo Illusion

- Linear perspective provides context
- Side lines seem to converge
- Top line seems farther away
  - But the retinal images of the red lines are equal!





Factors that Influence Perception

- **Perceptual set** (perceptual expectancy): the tendency to perceive things a certain way because previous experiences or expectations influence those perceptions

- **Top-down processing**: the use of preexisting knowledge to organize individual features into a unified whole

Perceptual Set



- What you see in the centre figures depends on the order in which you look at the figures:
  - If you scan from the left, see an old woman
  - If you scan from the right, see a woman's figure

Look at the drawing. What do you see?
Then look at the two pictures on the next slide.



Would you have interpreted the first drawing differently if you had viewed these images first?



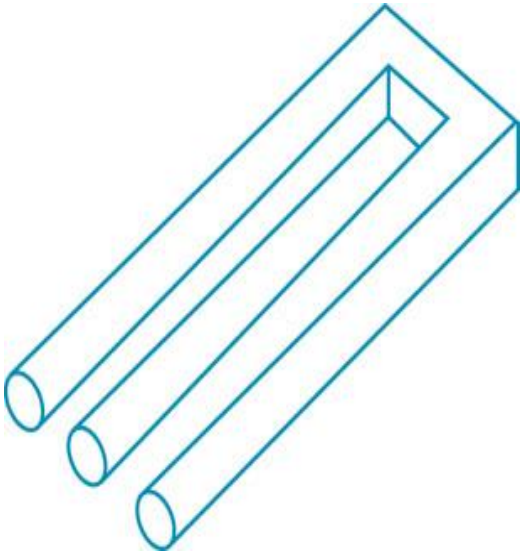
Old woman



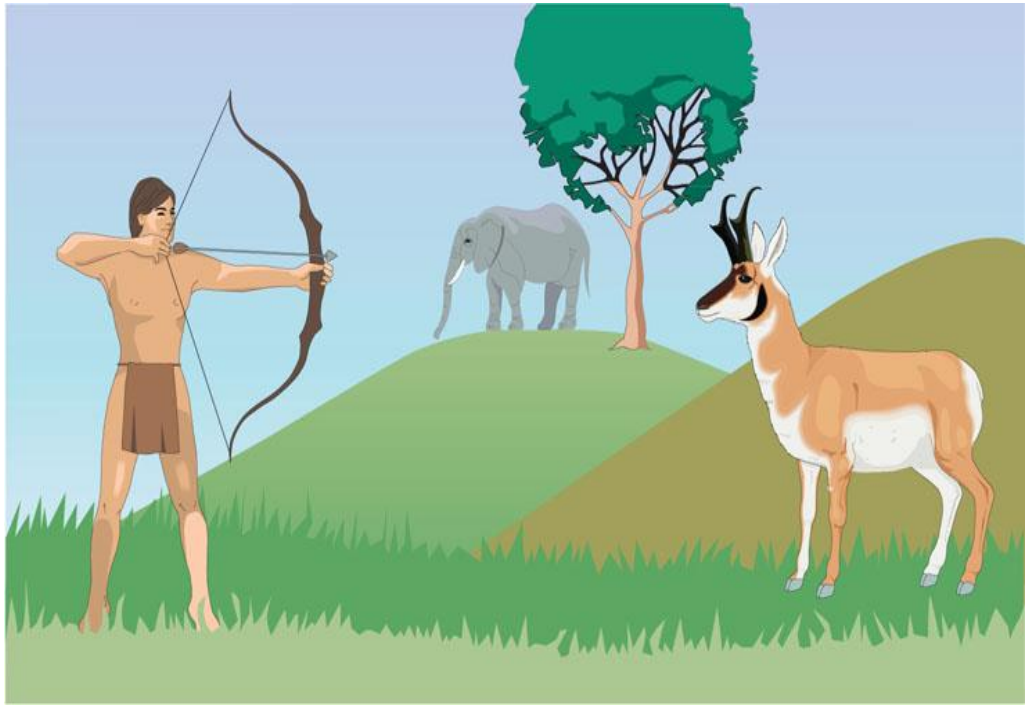
Young woman

Culture and Perception

- Cultural differences are reflected in depth perception
  - Zulu vs. Westerner perspectives



Two or three prongs?



The hunter is aiming at….?

Cultural Influences on Perception

- Muller lyer illusion – Zulu tribe in Africa is less susceptible to since they have few structures built using right angles and rectangles!

- Culture influences how we perceive the world.

Takeaways

- Sensation and perception are related but distinct processes:

sensation refers to the detection and initial encoding of physical stimuli by sensory receptors, whereas perception involves organising and interpreting this sensory information to create meaningful experience

- Perception is constructive rather than purely passive.

- The brain combines incoming sensory data with prior knowledge, expectations, context, and social-cultural experience. Consequently, people may encounter the same physical environment yet perceive it differently.

End